

P H A R M A L M S

Training Administrator Portal

Complete Master Specification

18 Screens · ADM-01 through ADM-18 · 360° Admin User Stories
FDA 21 CFR Part 11 · GMP Annex 11 · ALCOA+ · ICH Q10 · GAMP 5
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1. Admin Portal Design Principles & Tokens

The Training Administrator portal serves three admin personas: Training Administrator (org-wide training operations), Compliance Officer (regulatory oversight and audit readiness), and System Administrator (user, role, and system configuration). Design principles apply to all three with persona-specific rules noted.

1.1 Core Design Principles

Principle 1 — Authority with accountability

Admin actions have immediate and widespread impact. Assigning training to 10,000 employees, revoking a certificate, or triggering an org-wide retraining cascade cannot be easily undone. Every destructive or batch action must show a confirmation dialog with an exact count of affected records ("This will create 8,423 new training assignments. Proceed?"). All admin actions write to audit trail before confirmation is dismissed.

Principle 2 — Org-wide visibility at a glance

The admin portal sees the entire organisation. Compliance rate, overdue counts, HRIS sync status, and inspector session status must be visible within 3 seconds of login. The dashboard is the single source of truth for compliance readiness. Never require an admin to navigate 3 levels deep to answer "Are we compliant today?"

Principle 3 — Inspection readiness as a first-class feature

Every screen is designed so that data can be surfaced to an FDA/EMA inspector without modification. Audit packages can be generated in one click. Auditor session access can be provisioned and scoped in 60 seconds. The admin never needs to manually compile evidence — the system produces it.

Principle 4 — Role separation enforced in UI

Training Admin can assign, configure, and manage. QA Officer can approve, reject, and e-sign. System Admin can manage users and configuration. No single persona can perform the full compliance chain end-to-end (create course + approve course + assign to self). Separation of duties is enforced both in RBAC (server-side) and in UI (correct actions hidden from wrong role, not just disabled).

Principle 5 — Bulk operations with safety gates

Admins routinely operate at scale: bulk-assigning 500 employees, bulk-importing 200 new hires, bulk-revoking certs from a superseded SOP. Every bulk operation has three phases: (1) preview — show exactly what will change, (2) confirm — require explicit confirmation with affected count, (3) execute with progress indicator. Bulk operations are reversible where compliance permits.

Principle 6 — HRIS as the system of record for identity

Employee identity (name, email, department, job role, hire date, termination date) comes from HRIS. Admin cannot manually edit HRIS-managed fields — they are read-only with a "Managed by HRIS" badge. Manual edits risk divergence with the source of truth. HRIS sync status is always visible in the header so admin knows when data is fresh.

1.2 Design Tokens

Same semantic color system as the employee and trainer portals — consistent tokens across all portals.

TOKEN	USAGE (ADMIN PORTAL)
--em (#059669)	Compliant, approved, published, primary action button, active nav
--em2 (#D1FAE5)	Active nav bg, compliant badge bg, valid cert bg, selected row
--em4 (#047857)	Text on em2 badges, primary button hover, approved text
--red (#DC2626)	Overdue, failed, rejected, critical error, revoked cert, destructive action
--redbg (#FEF2F2)	Danger banner bg, rejection badge bg, revoked state bg
--amb (#D97706)	Under review, warning, pending QA, expiring certs, due-soon
--ambbg (#FFFBEB)	Pending badge bg, warning banner bg, due-soon indicator bg
--blu (#1D4ED8)	In progress, informational, version numbers, HRIS sync indicator
--blubg (#EFF6FF)	Info panels, metadata display, sync status bg
--ora (#C2410C)	CRITICAL SOP designation, retraining trigger, mandatory training badge
--orabg (#FFF7ED)	Critical SOP banner bg, mandatory badge bg
--pur (#6D28D9)	CAPA-linked training, waiver flags, inspector session active
--purbg (#F5F3FF)	CAPA badge bg, inspector session indicator bg
--teal (#0D9488)	Under review status, QA workflow state
--ink (#111827)	Page titles, table headings, high-emphasis text
--ink2 (#374151)	Body text, form labels, table data cells
--ink3 (#6B7280)	Metadata, timestamps, helper text, secondary labels
--bg (#F9FAFB)	Page background — never white for page level
--bg2 (#F3F4F6)	Table alternating rows, input fields, skeleton loaders
--bdr (#E5E7EB)	All borders, dividers, card outlines

1.3 Admin Portal App Shell

The admin shell wraps every authenticated admin screen. It uses a wider sidebar (256px) with section groupings that reflect the admin’s operational domains.

SECTION	NAV ITEMS
OVERVIEW	Dashboard (/admin), HRIS Sync Status, Compliance Summary
ORGANISATION	Users (/admin/users), Departments (/admin/org/departments), Sites (/admin/org/sites), Roles & Permissions (/admin/roles)
TRAINING	Training Matrix (/admin/training-matrix), Assignments (/admin/assignments), Training Waivers (/admin/waivers)
COMPLIANCE	Compliance Dashboard (/admin/compliance), Certificate Management (/admin/certificates), CAPA Management (/admin/capa)
CONTENT	SOP Documents (/admin/sop-documents), Course Library (read-only view)
AUDIT & REPORTING	Audit Log (/admin/audit-log), Reports (/admin/reports), Inspection Readiness (/admin/inspection)
SYSTEM	System Configuration (/admin/config), Notification Management (/admin/notifications), Scheduled Jobs (/admin/jobs)

1.4 Header Bar (all admin screens)

ELEMENT	SPECIFICATION
Org logo + name	Organisation logo (48px) + org name. Click -> admin dashboard.
HRIS sync badge	"HRIS Synced 14 min ago" green badge. Amber if >60 min. Red if >24 hrs or sync failed. Click -> HRIS sync log.
Inspector session indicator	"Inspector Active" amber pulsing badge when an auditor session is open. Click -> ADM-13 Inspection Readiness.
Search bar	Global admin search: employee name/ID, course title, SOP number, certificate ID. Max 500px.
Notification bell	Red badge when unread admin alerts: sync failures, DLQ depth > 0, SLA breach, CAPA overdue. Slide-out panel.
Avatar + role	36px circle + initials. Shows "Training Admin" "Compliance Officer" "System Admin" role badge. Click -> profile.

1.5 Typography Scale

LEVEL	SPECIFICATION
Page title (H1)	20px / 800 / ls -.5px / --ink. "User Management", "Compliance Dashboard"
Section heading (H2)	15px / 700 / ls -.2px / --ink. "Employee Directory", "Assignment Configuration"
Card heading (H3)	13px / 600 / --ink2. "HRIS Sync Status", "Matrix Changes"
Body text	13px / 400 / --ink2. Standard labels, table data, descriptions
Metadata	11px / 500 / --ink3. "Last synced: 14 min ago", "1 of 20,000 records"
Micro label	10px / 700 / UC / ls .07em / --ink4. "CRITICAL", "OVERDUE", "REVOKED"
Status badge	10px / 600 / pill. "ACTIVE", "SUSPENDED", "TERMINATED", "COMPLIANT"
Code / ID display	10px / monospace. SHA-256 hashes, employee IDs, cert UUIDs

2. ADM-01 — Admin Dashboard (360° Overview)

"As a Training Administrator, I want an org-wide overview of compliance status, pending actions, HRIS sync health, and inspection readiness — in a single screen loaded in under 3 seconds — so I can answer any compliance question without navigating multiple reports."

Route: /admin · Trigger: Login with admin/compliance officer/system admin role · Load target: <3s P95 · Priority: Phase 1 Critical

2.1 Zone 1 — Critical alerts banner row (topmost, non-dismissible)

ALERT TYPE	DISPLAY SPECIFICATION
HRIS sync failure	Red: "HRIS sync failed 3 hours ago. 47 employees may have incorrect assignments. View HRIS Log."
DLQ depth > 0	Red: "Dead-letter queue has 12 unprocessed events. Training assignments may be delayed. View Job Monitor."
Active inspector session	Amber: "FDA inspector session active — John Dawes (FDA Region II) · Expires in 3h 22m. View Session."
CAPA overdue	Orange: "3 CAPA training assignments are past their effectiveness check due date. View CAPAs."
SLA breach	Red: "14 employees are 7+ days overdue on mandatory GMP training. Escalation triggered."

2.2 Zone 2 — KPI stat cards (8 cards)

CARD	SPECIFICATION	PRIORITY
Total Employees	Count of active pharma_user records. Click -> User Management. Blue.	
Org Compliance Rate	AVG(individual compliance rates) across org. Emerald if >90%, amber 70-90%, red <70%. Formula shown as subtitle.	
Overdue Training	Count of enrollments past due date. Red if >0. Click -> filtered assignment list. +N increase badge.	
Expiring Certs (30d)	Certificates expiring within 30 days with no renewal started. Amber. Click -> certificate management.	

CARD	SPECIFICATION	PRIORITY
Pending QA	Course versions submitted for QA, awaiting review. Teal. Click -> course list filtered Under Review.	
Open Waivers	Training waivers pending admin or QA approval. Click -> waiver management. Amber.	
Open CAPAs	CAPA records with training required and not yet complete. Purple. Click -> CAPA management.	
HRIS Sync Health	Last sync time + result. Green: <60 min. Amber: 1-4 hrs. Red: >4 hrs or failed.	

2.3 Zone 3 — Department compliance heatmap

- Grid: rows = departments, columns = compliance rate ranges (0-60%, 60-70%, 70-80%, 80-90%, 90-100%)
- Each cell: count of employees in that department x compliance bracket
- Color: red cells (0-60%), amber (60-80%), emerald (90-100%)
- "Click any cell → filtered user list for that department + compliance range"
- "Export as PDF" button → generates watermarked compliance heatmap report

2.4 Zone 4 — Compliance trend (12-month sparkline)

- Line chart: 12 monthly org compliance rate data points from analytics_snapshot
- Emerald line with monthly markers. Baseline target line (configurable, default 90%)
- "Below target" regions shaded red. "Trend: +2.4% vs last month" badge
- Version selector: can overlay compliance trend for specific department or site

2.5 Zone 5 — Action queue (requires admin attention)

Three columns: Pending Waivers | Overdue Assignments (>7d) | CAPA Training Due

- Each item: employee name + course + days overdue/pending + action button
- "Assign Reviewer" | "Send Escalation" | "View CAPA" — inline actions without navigation

2.6 Zone 6 — Recent system activity feed

- Last 20 system events: HRIS sync, bulk assignment, SOP update, cert revocation, inspection package generated
- Each item: timestamp UTC + event type badge + description + actor (SYSTEM or admin name)
- "View full audit log" link → ADM-16

2.7 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
AdminDashboardLoaded	access_log	admin_id, load_time_ms, org_compliance_rate, overdue_count, pending_qa_count, timestamp

3. ADM-02 — User Management (Employee Directory)

"As a Training Administrator, I want to search, view, create, edit, suspend, and terminate employee accounts — with HRIS-managed fields protected from manual edits — so that user identity data stays consistent and training records are always attributed to the correct person."

Route: /admin/users · Priority: Phase 1 Critical

3.1 Employee directory table

- Search bar: full-text on name, email, employee ID, department. Results update on keystroke (300ms debounce).
- Filter row: Status (Active / Suspended / Terminated) | Department | Site | Job Role | MFA Enabled | Compliance Status (Compliant / Overdue)
- Sort: by name, compliance rate, hire date, last activity
- "Export Directory" button → CSV with SHA-256 hash in footer, AuditTrail: DirectoryExported

COLUMN	SPECIFICATION
Employee	Avatar (32px initials circle) + full name + employee ID (small). Click row → employee profile.
Department / Role	Department + job role below. "Managed by HRIS" tooltip on hover.
Site	Site name. Multi-site employees show primary site.
Hire date	ISO format. Used for training_matrix due_days_from_hire calculation.
Compliance %	Individual compliance rate ring (small). Color-coded by threshold.
Status	"ACTIVE" (green) "SUSPENDED" (amber) "TERMINATED" (red). Click → change status dialog.
MFA	Checkmark if user_mfa.mfa_enabled=true. Warning icon if not enrolled.
Last login	Relative timestamp. Red if >30 days (inactive user risk).
Actions	View Profile Reset MFA Suspend Terminate Impersonate (read-only).

3.2 Employee profile (drill-down)

- Route: /admin/users/:userId
- HRIS-managed fields (read-only, labeled "Managed by HRIS"): full name, email, employee ID, department, job role, site, hire date, manager ID
- Admin-editable fields: additional role assignments (RBAC), MFA reset, status change, preferred language

- Training overview panel: compliance rate ring, overdue count, active enrollments, cert portfolio health
- Active sessions panel: list of user_session rows with device/IP/last active. "Revoke all sessions" button (AuditTrail: AdminRevokedAllSessions)
- Training history shortcut: "View Training History" button → read-only view of TrainingHistoryV2 for this employee

3.3 Create employee (manual — when HRIS not connected)

- Modal: full name, email, employee ID, department (dropdown), job role (dropdown), site (dropdown), hire date, manager (search)
- "Invite to portal" checkbox: sends welcome email with temp password
- Triggers AUTO-01 onboarding: training matrix assignments created within 15 min

3.4 Suspend / Terminate actions

Termination compliance requirements
Termination must revoke ALL active user sessions within 60 seconds (§11.10(d))
Active enrollments are cancelled (cancellation_reason = "employee_terminated")
Active sessions written with end_reason = "admin_terminated" and ended_at = NOW()
AuditTrail: EmployeeTerminated + SessionsRevoked (separate rows)
Employee ID remains unique for 10 years post-termination (never reused)
Training records and certificates are RETAINED permanently for historical audit

3.5 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
EmployeeCreated	pharma_user	user_id, email, department_id, job_role_id, site_id, hire_date, created_by_id, created_at
EmployeeUpdated	pharma_user	user_id, changed_fields[], old_value_json, new_value_json, updated_by_id
EmployeeSuspended	pharma_user	user_id, suspended_by_id, suspended_at, reason, sessions_revoked
EmployeeTerminated	pharma_user	user_id, terminated_by_id, terminated_at, reason, enrollments_cancelled, sessions_revoked
AdminRevokedAllSessions	user_session	target_user_id, revoked_by_id, session_count, revoked_at, reason
MFAResetByAdmin	user_mfa	target_user_id, reset_by_id, reset_at, reason
DirectoryExported	report_export	exported_by_id, filter_params, record_count, file_hash, exported_at

4. ADM-03 — Bulk Import & HRIS Integration Monitor

"As a Training Administrator, I want to bulk-import employees via CSV and monitor HRIS sync status in real time, so that new hires receive training assignments within 15 minutes of account creation and terminations are processed within 60 seconds."

Route: /admin/users/import (bulk import) + /admin/integrations/hris (sync monitor) · Priority: Phase 1 Critical

4.1 Bulk CSV import

Upload phase

- Drag-drop zone: "Drag CSV file here or Browse". Accepts .csv only. Max 10,000 rows per file.
- Template download: "Download employee import template (CSV)" — includes all required headers with example data
- File validation (server-side on upload): column headers must match template exactly, employee IDs must be unique, email format validation, department/role must exist in system

Preview phase (before import executes)

- "Preview Import" shows: X rows to CREATE (green), Y rows to UPDATE (blue), Z rows with errors (red)
- Error table: row number + column + error message. Must resolve all errors before proceeding.
- Duplicate detection: warns if employee ID or email already exists ("Will update, not create new")
- "Proceed with Import" primary button — confirmation dialog showing exact counts

Import execution

- Progress indicator: "Importing 500 records... 243 complete (48%)" with cancel button
- Result summary: "Import complete: 487 created, 13 updated, 0 errors". Download error log if any.
- AUTO-01 triggers: training matrix assignments created for all new hires within 15 min
- AuditTrail: BulkImportCompleted (import_log table row created)

4.2 HRIS Integration Monitor

Sync status overview

FIELD	SPECIFICATION
Last sync timestamp	ISO 8601 UTC. Color coded: <1hr green, 1-4hr amber, >4hr red.
Sync source	Workday / SuccessFactors / BambooHR / Custom API — from system_configuration.

FIELD	SPECIFICATION
Records received	Total records in last sync payload. Breakdown: new/updated/terminated.
Records processed	Successfully processed vs failed. Failed count in red if > 0.
Sessions revoked	Count of sessions revoked in last sync (termination events). Must be > 0 if terminations occurred.
Assignments created	New training assignments generated by HRIS hire events in last sync.
Certs obsoleted	Certificates marked obsolete due to role transfer in last sync.

Sync history table

- Last 30 sync runs: timestamp + status (success/partial/failed) + records_received + records_failed + duration_seconds
- Click row → full hris_sync_log detail: exact records processed, error messages, retried events
- "Trigger Manual Sync" button (emergency use): AuditTrail: HrisSyncManuallyTriggered

4.3 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
BulkImportCompleted	import_log	import_type:employee, filename, record_count, success_count, failure_count, imported_by_id, imported_at
HrisSyncCompleted	hris_sync_log	sync_type, source_system, records_received, records_processed, records_failed, sessions_revoked, assignments_created, certs_obsoleted, status, completed_at
HrisSyncFailed	hris_sync_log	sync_type, error_message, failed_at, retry_count (actor=SYSTEM)
HrisSyncManuallyTriggered	hris_sync_log	triggered_by_id, triggered_at, reason

5. ADM-04 — Role & Permission Management

"As a System Administrator, I want to define roles, assign permissions, and link users to roles so that the principle of least privilege is enforced and no single user can perform the entire training approval chain."

Route: /admin/roles · Priority: Phase 1 Critical

5.1 Role list

- Table: role name + code + user count + permissions count + system role flag + actions
- "System roles" (employee, trainer, qa_officer, training_admin, system_admin, auditor) cannot be deleted — edit icon disabled with tooltip "System role — permissions can be modified but role cannot be deleted"
- "Create Custom Role" button: for org-specific roles (e.g. "Site Quality Lead")

5.2 Role detail / permissions editor

- Permission matrix: resource column vs action column (create/read/update/delete/approve/export)
- Checkboxes per cell. System roles show pre-checked, non-editable cells in gray.
- Resources: organization, training, compliance, quality_event, audit, user, system_config, report, waiver, certificate
- "Separation of duties" guard: if trainer:approve is checked, qa_officer:approve is automatically forced off for that user (cannot hold both)

5.3 User-Role assignment

- "Assign Role" from user profile or bulk from role detail page
- Role assignment requires reason (free text). Stored in user_role with assigned_by_id + assigned_at + reason
- "Assign User Role" button → search employee + select role + enter reason + confirm

5.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
RoleCreated	role	role_id, name, code, created_by_id, created_at
RolePermissionUpdated	permission	role_id, resource, action, old_value:bool, new_value:bool, updated_by_id

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
UserRoleAssigned	user_role	user_id, role_id, assigned_by_id, assigned_at, reason
UserRoleRevoked	user_role	user_id, role_id, revoked_by_id, revoked_at, reason

6. ADM-05 — Organisation, Site & Department Management

"As a Training Administrator, I want to manage the org hierarchy (sites, departments, job roles) so that training matrix rules correctly resolve to the right employees and compliance metrics aggregate correctly."

Route: /admin/org · Priority: Phase 1 High

6.1 Org hierarchy tree

- Tree view: Organisation → Sites → Departments → Job Roles
- Expand/collapse nodes. Employee count at each level.
- Click a node → side panel with name, code, timezone, GMP classification, employee count, compliance rate

6.2 Site management

ACTION	SPECIFICATION
Create site	Name, code (unique), timezone (dropdown, default UTC), GMP classification (GMP / Non-GMP / Mixed)
Edit site	All fields editable. Code change requires confirmation ("Affects 847 employees and training matrix rules").
Archive site	All active employees must be transferred to another site first. Cannot archive with active employees.
Site compliance	Per-site compliance rate visible in site detail panel. Click → site-scoped compliance dashboard.

6.3 Job role management

- Job role name + code + department assignment + linked training matrix courses (count badge)
- "View Training Matrix" shortcut from job role row → opens ADM-08 filtered to this role
- Training matrix preview inline: "This role requires 8 mandatory courses + 3 optional"

6.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
SiteCreated	site	site_id, name, code, timezone, created_by_id
DepartmentCreated	department	dept_id, site_id, name, code, created_by_id
JobRoleCreated	job_role	role_id, dept_id, name, code, created_by_id
OrgStructureUpdated	organization	entity_type, entity_id, changed_fields[], old/new values, updated_by_id

7. ADM-06 — Training Assignment Management

"As a Training Administrator, I want to assign training to individuals, departments, job roles, or the entire organisation — with due dates, priorities, and mandatory/optional flags — so that the right employees are trained on the right content at the right time."

Route: /admin/assignments · Priority: Phase 1 Critical

7.1 Assignment list

- Filters: Status (active/cancelled/completed) | Target type (individual/dept/role/site-wide) | Due date range | Course | Priority | Overdue flag
- Table columns: assignment ID + target + course title + version + due date + priority + status + enrolled count + completion % + actions
- "Export Assignments" CSV with AuditTrail: ReportExported
- Bulk cancel: select multiple → "Cancel Selected" → mandatory cancellation reason

7.2 Create Assignment (wizard)

Step 1 — Select target

- Target type radio: Individual | Department | Job Role | Site | Org-wide
- If Individual: employee search by name/ID. Multi-select up to 200.
- If Department/Role/Site: dropdown selector with employee count preview ("This will assign to 1,247 employees")
- Duplicate check: "47 of these employees already have an active enrollment for this course. They will be excluded."

Step 2 — Select course

- Course search (QA Approved / Published only). Shows: title + version + current avg pass rate + enrolled count.
- Version selection: defaults to latest published. Can select specific version.

Step 3 — Configure assignment

FIELD	SPECIFICATION
Due date	Date picker. Cannot be in the past. "Due in X days" relative label.
Priority	High (red) / Medium (amber, default) / Low (gray). Affects sort order on employee dashboard.

FIELD	SPECIFICATION
Reason / note	Free text. Shown to employee in assignment notification and training list. Max 500 chars.
Mandatory / Optional	Toggle. Mandatory = counts against compliance rate. Optional = visible in catalogue but excluded from compliance calc.
Assignment source	Auto-filled: manual / matrix / sop_update / capa / hris_hire. Non-editable from UI (set by system for automated flows).

Step 4 — Confirm & Execute

- "Create 1,200 training assignments for GMP Fundamentals v2.0 due 15 April 2026. Confirm?"
- Progress indicator during bulk create. Notification dispatched when complete.
- AuditTrail: TrainingAssigned (batch) with assignment_count in key fields

7.3 Assignment detail & cancel

- Route: /admin/assignments/:assignmentId
- Read-only fields: target, course, version, due date, reason, enrolled count, completion %
- "Cancel Assignment" red button: requires mandatory cancellation reason. All in_progress and not_started enrollments set to status=cancelled.
- AuditTrail: TrainingAssignmentCancelled with cancellation_reason and affected_enrollment_count

7.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
TrainingAssigned	training_assignment (batch)	target_type, course_version_id, due_date, priority, reason, mandatory:bool, assignment_count, assigned_by_id, assigned_at
TrainingAssignmentCancelled	training_assignment	assignment_id, cancelled_by_id, cancelled_at, cancellation_reason, affected_enrollment_count
AssignmentDueDateExtended	training_assignment	assignment_id, old_due_date, new_due_date, extended_by_id, reason

8. ADM-07 — Training Waiver Management

"As a Training Administrator, I want to request, review, approve, and track training waivers so that employees with documented prior learning or equivalent experience can be marked compliant without repeating already-mastered training."

Route: /admin/waivers · Priority: Phase 1 High

8.1 Waiver list

- Filters: Status (pending / approved / rejected / expired) | Employee | Course | Requested by | Date range

COLUMN	SPECIFICATION
Employee	Name + ID. Click → employee profile.
Course / Version	Course title + version for which waiver was requested.
Requested by	Admin who initiated. Trainers cannot self-request waivers — Training Admin only.
Request date	ISO UTC timestamp of request.
Status	PENDING (amber) APPROVED (green) REJECTED (red) EXPIRED (gray, past expiry date).
Expiry date	Date waiver expires. After expiry, employee must complete training or new waiver required.
Actions	"Review" (pending) "View" (approved/rejected) "Revoke" (approved) "Extend" (approved).

8.2 Create waiver request

- Employee search + course selection
- Reason: free text, required. "Employee holds equivalent ISO 9001 Lead Auditor certification."
- Evidence: file upload (PDF/image). Stored in S3, SHA-256 hash computed. evidence_storage_path on training_waiver table.
- Waiver expiry date: defaults to 12 months from request date. Admin can override.
- QA e-signature required for approval (Training Admin initiates, QA Officer approves)

8.3 Approve / Reject waiver (QA Officer action)

- Review pane: employee details, course, reason, evidence (PDF preview or download link)

- "Approve" → QA e-sign flow (same as course approval). AuditTrail: WaiverApproved + EsignatureCreated
- "Reject" → mandatory rejection reason field. Employee notified. AuditTrail: WaiverRejected
- Approved waiver: training_waiver.status = approved. Employee compliance calc counts waiver as COMPLIANT.

8.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
WaiverRequested	training_waiver	waiver_id, user_id, course_id, reason, evidence_path, evidence_hash, requested_by_id, expires_at
WaiverApproved	training_waiver	waiver_id, approved_by_id, approved_at, esignature_id, comments §11.50
WaiverRejected	training_waiver	waiver_id, rejected_by_id, rejected_at, rejection_reason
WaiverRevoked	training_waiver	waiver_id, revoked_by_id, revoked_at, reason (e.g. new SOP version)
WaiverExtended	training_waiver	waiver_id, old_expires_at, new_expires_at, extended_by_id, reason

9. ADM-08 — Training Matrix Manager

"As a Training Administrator, I want to define and version the training matrix — mapping job roles to required courses with due-day windows and retraining intervals — so that new employees are automatically assigned the correct training based on their role from day one."

Route: /admin/training-matrix · Priority: Phase 1 Critical

9.1 Matrix grid view

- Rows = Job roles (grouped by department). Columns = Courses.
- Cell values: Required (emerald) | Optional (blue) | Not Required (empty). Click cell → configure.
- Cell configuration: is_mandatory, due_days_from_hire, retraining_interval_days
- Filter: by department, site, mandatory-only toggle, course category
- "Compare with previous version" toggle: shows diff overlay (green = added, red = removed, amber = modified)

9.2 Matrix edit flow

Change preview before save

- "You are modifying 14 matrix cells affecting 3,240 employees. Changes will take effect after QA e-signature." — confirmation required
- Diff table: role + course + old value + new value. Admin must review all changes before signing.

QA e-signature gate

Training matrix changes require QA e-signature (21 CFR Part 11)
Matrix changes affect what training is mandatory for thousands of employees
A QA Officer must e-sign before any matrix change takes effect
E-signature uses the "training_matrix_approval" meaning from SignatureMeaning table
AuditTrail: TrainingMatrixUpdated with qa_esignature_id and diff_json
After signing, new matrix is effective immediately — HRIS-triggered assignments use new rules

9.3 Matrix version history

- Every approved matrix version stored with: version number, effective date, approved_by, esignature_id
- "View matrix as-of [date]" feature: renders grid as it was on any historical date

- Used in audit to answer: "What was the required training for Quality Control Inspector on 15 Jan 2025?"

9.4 Bulk CSV import

- Same 3-phase flow as user import: Upload → Preview diff → QA e-sign → Execute
- Template download with all column headers. Validation: role and course must exist.

9.5 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
TrainingMatrixUpdated	training_matrix	changed_cells_json (array of {role_id, course_id, old_config, new_config}), diff_preview_json, updated_by_id, qa_esignature_id, effective_at
TrainingMatrixVersionCreated	training_matrix	version_number, base_version_id, created_by_id, created_at, change_notes
TrainingMatrixApproved	training_matrix	version_number, approved_by_id, approved_at, esignature_id §11.50

10. ADM-09 — Certificate Management & Revocation

"As a Training Administrator, I want to view, search, and manage all certificates — including revoking certificates where training integrity was compromised — so that employees never hold invalid credentials in regulated processes."

Route: /admin/certificates · Priority: Phase 1 High

10.1 Certificate directory

- Search: employee name/ID, course title, certificate ID (UUID), SOP number
- Filters: Status (active/expiring/expired/obsolete/revoked) | Course | Department | Expiry date range | Version

COLUMN	SPECIFICATION
Employee	Name + ID. Click → employee profile.
Course / Version	Course title + version trained on. Shows if superseded.
Issued date	ISO UTC. Links to training_record and electronic_signature.
Expiry date	Date or "No expiry". Color-coded: >60d blue, 31-60d amber, <30d red, expired red pulsing.
Status	ACTIVE EXPIRING_SOON EXPIRED OBSOLETE REVOKED. Never hidden.
E-signature	"Verified" or "Integrity violation" badge. HMAC re-checked on load.
Actions	"View Certificate" "Verify Integrity" "Revoke" (active only) "Download PDF"

10.2 Certificate revocation

Revocation compliance requirements
Certificate revocation is an irreversible action — once revoked, cannot be un-revoked
Revocation requires mandatory reason (dropdown: training_integrity_failure / document_update / error / disciplinary)
QA e-signature required for revocation (QA Officer must sign)
Revocation immediately removes certificate from employee's active credentials
Employee notified automatically: "Your certificate for [Course] has been revoked by QA on [date]. Reason: [reason]."

Persistent red banner on employee dashboard — cannot be dismissed
New enrollment created automatically if revocation reason = training_integrity_failure

10.3 Batch certificate operations

- "Mark Obsolete" batch: select certs from a specific course version → mark all as OBSOLETE when new version published (usually automated via AUTO-03, but manual override available)
- "Export Certificate Report" → CSV with employee name, course, version, issued date, expiry, status. SHA-256 hash in footer.
- "QR Verification Test" → admin can scan any certificate QR and verify the /verify/:token response. Tests that public verification is working.

10.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
CertificateRevoked	certificate	cert_id, revoked_by_id, revoked_at, reason, esignature_id, user_notified:bool §11.50
CertificateMarkedObsolete	certificate	cert_id, obsoleted_by_id, obsoleted_at, reason (actor:SYSTEM for AUTO-03)
CertificateIntegrityChecked	certificate	cert_id, checked_by_id, checked_at, result:verified violation
CertificateReportExported	report_export	exported_by_id, filter_params, record_count, file_hash, exported_at

11. ADM-10 — Compliance Dashboard (Org-Wide)

"As a Compliance Officer, I want a single dashboard showing compliance rates by department, site, and job role — with drill-down to individual employees — so I can identify compliance gaps before an FDA inspection and take corrective action."

Route: /admin/compliance · Priority: Phase 1 Critical

11.1 Org-level compliance summary

- Compliance rate ring (large, 120px): current org-wide rate. Formula: COUNT(completed+waived) / COUNT(assigned where due<=today)
- "Target: 95% | Current: 87.3% | Gap: 7.7% | Affected: 1,540 employees" — gap analysis strip
- Trend chart: 12-month compliance rate line. Alert line at configured SLA threshold (default 90%).
- "Inspection ready" indicator: green checkmark if compliance > threshold + no open critical findings. Red cross if either condition fails.

11.2 Department compliance breakdown

COLUMN	SPECIFICATION
Department	Name + site. Click → department-scoped compliance detail.
Employees	Total active headcount.
Compliant	Count with compliance rate >= 100%. Emerald.
Overdue	Count with at least 1 overdue assignment. Red if > 0.
Compliance %	Progress bar with %. Color-coded by threshold.
Trend	Up/down arrow with +/- delta vs last month.
Actions	"Assign Missing Training" "Export Report" "View Employees"

11.3 Individual non-compliance list

- Table of employees with compliance rate < 100%, sorted by worst first
- Columns: employee + department + compliance % + overdue count + courses overdue (list)
- "Send Reminder" button: dispatches escalation notification. AuditTrail: EscalationDispatched
- "Assign Training" button: shortcut to ADM-06 pre-filled with this employee

11.4 SOP compliance matrix

- Grid: rows = SOPs (linked to courses), columns = departments. Cells = % employees with valid cert for that SOP.
- Click cell → list of employees in that department who are NOT certified on that SOP
- "Most problematic SOP" highlight: SOP with lowest avg certification across all departments

11.5 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
ComplianceDashboardViewed	access_log	admin_id, scope (org/dept/site), viewed_at, compliance_rate_at_view
ComplianceReportExported	report_export	exported_by_id, scope, filter_params, record_count, file_hash, exported_at
EscalationDispatched	notification	target_user_id, dispatched_by_id, dispatched_at, overdue_count, notification_type:escalation

12. ADM-11 — SOP Document Management

"As a Training Administrator, I want to upload, version, and publish SOP documents — and automatically trigger retraining when a new version is published — so that training always reflects the current controlled document version."

Route: /admin/sop-documents · Priority: Phase 1 Critical

12.1 SOP library list

- Filters: Status (active/superseded/draft) | Criticality (CRITICAL/Standard) | Department scope | Effective date range

COLUMN	SPECIFICATION
SOP number	Unique document number (e.g. "SOP-101"). Click → SOP detail.
Title	Full SOP title. Linked to course count badge.
Current version	Version string (e.g. "v4.2"). Click → version history.
Criticality	CRITICAL (orange badge) Standard (gray badge). Determines retraining deadline (14d vs 30d).
Effective date	Date current version came into effect.
Linked courses	Count of course versions linked to this SOP. Click → course list filtered by SOP.
Employee certs	Count of active employee certificates covering this SOP.
Actions	"View" "Upload new version" "Set criticality" "View retraining impact"

12.2 Upload new SOP version

- Same upload flow as TRN-02 material upload: drag-drop, SHA-256 hash, AV scan, MIME validation
- Version number: auto-incremented (major.minor). Admin can override to major version (triggers isMajorVersion=true)
- Change summary textarea (required): "Updated to reflect revised aseptic gowning procedure per EU GMP Annex 1 2023." This text appears in employee SOP retraining acknowledgement modal (SCR-18).
- Affected roles selection: multi-select job roles. Determines retraining scope.
- Affected departments selection: multi-select or "All departments".
- "Preview retraining impact" before publish: shows count of employees who will be assigned retraining and how many active certs will be obsoleted.

12.3 Publish new version and retraining cascade

SOP publication triggers automatic retraining (AUTO-03)
On publish: document_version.status = effective. isMajorVersion flag checked.
System identifies all employees with certificate linked to old SOP version
Old certificates marked OBSOLETE (cert.status = obsolete, obsoleted_at = NOW())
New training_assignment rows created for all affected employees
Due date: NOW() + 14 days for CRITICAL SOPs NOW() + 30 days for Standard SOPs
Notifications dispatched: orange SOP retraining banner on employee dashboard
AuditTrail: RetrainingTriggeredFromSOP with affected_employee_count (actor=SYSTEM)
TrainingImpactAssessment row created for audit evidence of who was affected

12.4 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
SOPDocumentUploaded	document_version	document_id, version, file_hash, is_major_version, uploaded_by_id, change_summary
SOPDocumentPublished	document_version	document_id, version, effective_date, published_by_id, affected_role_ids, affected_dept_ids
RetrainingTriggeredFromSOP	enrollment (batch)	sop_document_id, new_version, affected_employee_count, certs_obsoleted, assignments_created, triggered_at (actor=SYSTEM)
TrainingImpactAssessment Created	training_impact_assessment	document_version_id, affected_count, obsoleted_cert_count, assignments_created, triggered_at

13. ADM-12 — CAPA Management (ICH Q10)

"As a Compliance Officer, I want to link corrective and preventive actions to mandatory training assignments so that training completion is required evidence before a CAPA can be closed, satisfying ICH Q10 training effectiveness requirements."

Route: /admin/capa · Priority: Phase 1 High

13.1 CAPA list

- Filters: Status (initiation/investigation/rca/training/effectiveness_check/closed) | Type (deviation/capa/change_control) | Site | Date range | Training completion flag

COLUMN	SPECIFICATION
CAPA reference	Quality event reference ID + type badge. Links to quality_event table.
Title	CAPA description title.
Status	INITIATION → INVESTIGATION → RCA → TRAINING REQUIRED → EFFECTIVENESS CHECK → CLOSED
Training required	"Yes" badge (purple) with course title if training_required=true. "None" if false.
Training completion	For CAPAs with training: X% complete badge. Blocks effectiveness check.
Effectiveness due	effectivenessCheckDue date. Red if overdue.
Actions	"View" "Link Training" "Complete RCA" "Close CAPA"

13.2 CAPA detail view

Quality event linkage

- qualityEvent.eventType: deviation | capa | change_control
- qualityEvent.referenceId: external QMS reference ID (e.g. Veeva QMS ID)
- rootCause text area: free text RCA. rcaCompletedAt timestamp.

Training requirement linkage

- "Requires training" toggle: links CAPA to a training assignment.
- "Link existing assignment" OR "Create new assignment" shortcut to ADM-06
- Once all linked assignment employees complete training: CAPA automatically moves to EFFECTIVENESS_CHECK stage (AUTO-08)

Effectiveness check

- effectivenessCheckDue date: admin-set, typically 90 days post-training completion
- At effectiveness check date: admin documents evidence of training effectiveness
- Optional: link to training_effectiveness_survey results for Kirkpatrick Level 1-2 evidence
- "Close CAPA" button: QA e-signature required. AuditTrail: CapaClosed + EsignatureCreated

13.3 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
CapaCreated	capa	capa_id, quality_event_id, title, training_required:bool, created_by_id
CapaTrainingLinked	capa	capa_id, training_assignment_id, linked_by_id, linked_at
CapaMovedToEffectivenessCheck	capa	capa_id, all_training_completed_at (actor=SYSTEM, AUTO-08)
CapaClosed	capa	capa_id, closed_by_id, closed_at, effectiveness_evidence_text, esignature_id §11.50

14. ADM-13 — Inspection Readiness & Audit Package Generator

"As a Compliance Officer, I want to generate scoped inspection packages, provision time-limited read-only auditor access, and verify system integrity — in under 60 seconds — so that FDA/EMA inspectors can access exactly what they need without risk of data modification."

Route: /admin/inspection · Priority: Phase 1 Critical

14.1 Inspection readiness status panel

INDICATOR	DISPLAY SPECIFICATION
Audit trail integrity	Last SYS-WF-08 run result: "Passed — 0 hash mismatches, 0 sequence gaps" (green) or "FAILED" (red alert). Last run timestamp.
E-signature integrity	Count of certificates with HMAC verification status. "All 14,203 signatures verified" or count of violations (red).
Retention compliance	Confirms retention_policy rules are active and last archive job ran successfully.
System validation status	GAMP 5 / CSA validation package reference and last review date.
Open inspector sessions	Count of active auditor_session rows. Each with expiry time and access scope.

14.2 Provision auditor access

Auditor session configuration

FIELD	SPECIFICATION
Access type	Internal External FDA External EMA Customer audit. Determines scope defaults.
Auditor name(s)	Free text. Stored in inspection_record.inspector_names.
Scope: date range	From/to date range. Only records within this range are visible to auditor.
Scope: sites	Multi-select sites. Auditor sees only training data for selected sites.
Scope: departments	Multi-select departments. Narrows further if needed.

FIELD	SPECIFICATION
Session duration	Hours until token expiry. Max 72 hours. Default: 8 hours.
Access token delivery	Displayed once on screen + emailed to auditor. QR code for scanner-based access.

- "Provision Access" button → creates auditor_session + inspection_record rows. AuditTrail: InspectorSessionProvisioned
- Auditor token is single-use for session creation (never repeatable). Session itself then authenticated per call.
- "Revoke Session" red button: immediately invalidates token. AuditTrail: InspectorSessionRevoked.

14.3 Inspection package generator

- Package scope: date range + sites + departments + event types (training records, certificates, audit trail, e-signatures, SOP links)
- "Generate Package" → async job: bundles all records as paginated JSON + summary PDF with watermark
- Package SHA-256 hash computed over entire bundle. Stored in inspection_package.file_hash.
- "Official package" toggle: requires QA Officer e-signature. isOfficial=true flag. AuditTrail: InspectionPackageGenerated + EsignatureCreated.
- Package stored in S3 with lifecycle rule: accessible for 90 days then archived.
- Previous packages listed with: generated date + scope description + generated by + file hash + download link

14.4 Auditor activity log

- Real-time feed of auditor_page_log rows: every page the auditor viewed, entity type, entity ID, time on page, exported flag
- "Stop session" button → AuditTrail: InspectorSessionEnded (admin-initiated)

14.5 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
InspectorSessionProvisioned	auditor_session	session_id, access_type, auditor_names, scope_start, scope_end, scope_sites, scope_depts, expires_at, provisioned_by_id
InspectorSessionRevoked	auditor_session	session_id, revoked_by_id, revoked_at, reason
InspectorSessionEnded	auditor_session	session_id, ended_at, pages_viewed_count, duration_seconds, ended_reason
InspectionPackageGenerated	inspection_package	package_id, scope_description, record_count, file_hash, storage_url, is_official, esignature_id, generated_by_id

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
AuditIntegrityCheckRun	audit_integrity_result	records_checked, hash_mismatches, sequence_gaps, result:passed failed, checked_at (actor=SYSTEM)

15. ADM-14 — System Configuration

"As a System Administrator, I want to manage org-level configuration settings — session timeouts, password policies, notification preferences, feature flags — so that the system behaviour matches the validated configuration without requiring code changes."

Route: /admin/config · Priority: Phase 1 High

15.1 Configuration categories

Security policy (21 CFR §11.300 settings)

SETTING	SPECIFICATION
Session idle timeout	Minutes until session timeout warning (SCR-22). Default: 15. Range: 5-60. Non-zero required.
Max failed login attempts	Attempts before account lock. Default: 5. Range: 3-10.
Password minimum length	Characters. Default: 12. Minimum: 8 per FDA guidance.
Password complexity	Checkboxes: uppercase, lowercase, number, special char required.
Password history	Number of previous passwords to prevent reuse. Default: 12.
MFA enforcement	Dropdown: Optional Required for all Required for admins only Required for QA only.
Concurrent sessions	Max active sessions per user. Default: 1. Range: 1-5.

Compliance policy settings

SETTING	SPECIFICATION
Organisation compliance target	Target compliance rate % for SLA monitoring and dashboard. Default: 90%.
Standard SOP retraining deadline	Days from SOP publish to retraining due. Default: 30. Affects AUTO-03.
Critical SOP retraining deadline	Days from critical SOP publish. Default: 14. Affects AUTO-03.
Cert expiry warning thresholds	Days before expiry to show 90d/60d/30d/7d banners on employee dashboard (AUTO-05).

SETTING	SPECIFICATION
E-sign window	Seconds for employee to sign training record after passing assessment. Default: 60.
Assessment max attempts (default)	Default max attempts for new assessments. Trainer can override per assessment. Default: 3.

Feature flags

- Feature flags per org: AI_QUESTION_GENERATION | GAMIFICATION | CLASSROOM_TRAINING | EXTERNAL_TRAINING | SCORM | XAPI | HRIS_INTEGRATION | INSPECTION_PACKAGE | TRAINING_EFFECTIVENESS_SURVEY
- Each flag: on/off toggle + description. Some flags require QA e-signature to change (noted in UI).

15.2 Configuration change audit

All configuration changes are audited and require QA e-signature for security-critical settings
Security policy changes (session timeout, MFA enforcement, password policy) require QA e-signature before taking effect
Feature flag changes that affect the audit trail (e.g. disabling XAPI statements) require QA e-signature
AuditTrail: SystemConfigChanged with old_value and new_value for every changed key
Configuration is versioned: admin can view configuration as-it-was on any date

15.3 Events & AuditTrail

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
SystemConfigChanged	system_configuration	key, old_value, new_value, changed_by_id, changed_at, esignature_id (for security-critical keys)
FeatureFlagToggled	feature_flag	key, old_enabled:bool, new_enabled:bool, toggled_by_id, toggled_at, esignature_id (if required)

16. Additional Admin Screens (ADM-15 to ADM-18)

Beyond the core screens, a complete Admin Portal requires the following supporting screens. All follow the design principles and token system from Part 1.

ADM-15 — Analytics & Reporting Hub

Route: /admin/reports · Priority: Phase 1 High

"As a Training Administrator, I want to generate, schedule, and export compliance and training analytics reports with cryptographic proof of integrity, so that I can provide evidence to management and regulators without manually compiling data."

- Report library: pre-built report definitions covering: compliance summary, training completion, certificate portfolio, overdue analysis, question difficulty, engagement quality, retraining velocity, SLA breach history, HRIS sync log
- Custom report builder: drag-and-drop field selector from approved analytics tables. Saved as report_definition.
- Scheduled reports: configure recurring export (daily/weekly/monthly). Delivered via email or to S3 path.
- All exports: SHA-256 hash in footer, watermark with admin name + timestamp, AuditTrail: AnalyticsReportExported
- SLA breach history: chart of org compliance rate vs SLA threshold over time. Annotations on breach events.

ADM-16 — Full System Audit Log Viewer

Route: /admin/audit-log · Priority: Phase 2 High

"As a Compliance Officer, I want read-only access to the complete system audit trail with filtering, pagination, and export — and the ability to verify SHA-256 row hashes inline — so that I can respond to any inspector question within minutes."

- Access to FULL audit_trail table (unlike trainer scope which is own-courses only)
- Filters: date range (calendar picker, ISO UTC), action type (multi-select), entity type (multi-select), actor (employee search), entity ID (direct lookup)
- Each row: timestamp UTC | sequence ID | action | entity type | entity ID | actor | key fields JSON | SHA-256 row hash | "Verify hash" button
- "Verify hash" button: re-computes SHA-256 of row content server-side and compares. Shows VERIFIED (green) or INTEGRITY VIOLATION (red alert + QA notification dispatched).
- Export: paginated CSV (max 100k rows per export). AuditTrail: AuditLogExported (file_hash, filter_params, record_count)

- Sequence gap detector: button "Check for sequence gaps" → calls SYS-WF-08 on demand. Shows any missing IDs in the sequence.

ADM-17 — Notification Management

Route: /admin/notifications · Priority: Phase 2 Medium

"As a Training Administrator, I want to manage notification templates, view delivery logs, and control the escalation ladder so that employees receive timely, compliant reminders without flooding inboxes."

- Notification log: table of all notification rows with delivery status (sent/failed/bounced), channel (email/push/in_app), timestamp, external message ID (e.g. SendGrid ID)
- Delivery failure alert: red badge when notification_log has status=failed in last 24h. Click → filtered log view.
- Escalation ladder configuration: define the notification schedule for overdue training (Day -30, -14, -7, -3, 0, +1, +3, +7, +14). Recipients at each level (employee only, then +manager, then +dept head, then +site head).
- "Send test notification" button: dispatches a test notification to the admin's own account. Does not write AuditTrail (test flag set).
- Quiet hours global config: hours when non-urgent notifications are suppressed org-wide.

ADM-18 — Scheduled Jobs Monitor

Route: /admin/jobs · Priority: Phase 2 Medium

"As a System Administrator, I want to monitor background job execution, view errors, and trigger manual reruns so that I can ensure compliance-critical automation (cert expiry checks, compliance calculations, HRIS sync) is running reliably."

JOB NAME	SCHEDULE + PURPOSE
CertExpiryCheck (AUTO-05)	Runs nightly 02:00 UTC. Scans certificate.expires_at for 90/60/30/7d thresholds. Creates notifications.
ComplianceCalc (AUTO-07)	Runs nightly 01:00 UTC. Computes analytics_snapshot for all orgs and departments.
NotificationWorker (AUTO-06)	Runs every 15 min. Evaluates escalation ladder, dispatches notifications.
AuditIntegrityCheck (SYS-WF-08)	Runs nightly 03:00 UTC. Verifies SHA-256 row hashes + sequence gaps on audit_trail.
RetentionArchiver	Runs monthly. Applies retention_policy rules, archives to retention_archive.
HrisSyncJob	Frequency configurable. Processes HRIS event queue.

- Job status table: last run start + end + status + records_processed + records_affected + error_details
- "Rerun job" button (emergency): AuditTrail: ScheduledJobManuallyTriggered (triggered_by_id + reason)
- "DLQ" panel: dead_letter_queue rows with failure_reason + retry count. "Retry" and "Mark resolved" buttons.
- Alert: red banner if any job has status=failed in last 24h or DLQ depth > 0

17. Complete AuditTrail Reference — Admin Portal

Every AuditTrail INSERT for the Admin Portal. All entries are append-only. No UPDATE or DELETE ever permitted. Row-level SHA-256 hash stored. ID sequence monitored nightly. actor=SYSTEM for all automated events.

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
AdminDashboardLoaded	access_log	admin_id, org_compliance_rate, overdue_count, pending_qa_count, timestamp
EmployeeCreated	pharma_user	user_id, email, department_id, job_role_id, hire_date, created_by_id
EmployeeUpdated	pharma_user	user_id, changed_fields[], old_value_json, new_value_json, updated_by_id
EmployeeSuspended	pharma_user	user_id, suspended_by_id, reason, sessions_revoked, suspended_at
EmployeeTerminated	pharma_user	user_id, terminated_by_id, reason, enrollments_cancelled, sessions_revoked · §11.10(d)
AdminRevokedAllSessions	user_session	target_user_id, revoked_by_id, session_count, revoked_at, reason
MFAResetByAdmin	user_mfa	target_user_id, reset_by_id, reset_at, reason
DirectoryExported	report_export	exported_by_id, filter_params, record_count, file_hash · §11.10(b)
BulkImportCompleted	import_log	import_type, filename, record_count, success_count, failure_count, imported_by_id
HrisSyncCompleted	hris_sync_log	source_system, records_received, processed, failed, sessions_revoked, assignments_created (actor=SYSTEM)
HrisSyncFailed	hris_sync_log	error_message, failed_at, retry_count (actor=SYSTEM)
HrisSyncManuallyTriggered	hris_sync_log	triggered_by_id, triggered_at, reason
RoleCreated	role	role_id, name, code, created_by_id
RolePermissionUpdated	permission	role_id, resource, action, old_value, new_value, updated_by_id
UserRoleAssigned	user_role	user_id, role_id, assigned_by_id, reason, assigned_at
UserRoleRevoked	user_role	user_id, role_id, revoked_by_id, reason, revoked_at

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
SiteCreated	site	site_id, name, code, timezone, created_by_id
DepartmentCreated	department	dept_id, site_id, name, code, created_by_id
OrgStructureUpdated	organization	entity_type, entity_id, changed_fields[], old/new values, updated_by_id
TrainingAssigned	training_assignment (batch)	target_type, course_version_id, due_date, mandatory:bool, assignment_count, assigned_by_id
TrainingAssignmentCancelled	training_assignment	assignment_id, cancelled_by_id, reason, affected_enrollment_count
AssignmentDueDateExtended	training_assignment	assignment_id, old_due_date, new_due_date, extended_by_id, reason
WaiverRequested	training_waiver	waiver_id, user_id, course_id, reason, evidence_hash, requested_by_id, expires_at
WaiverApproved	training_waiver	waiver_id, approved_by_id, esignature_id, comments · §11.50
WaiverRejected	training_waiver	waiver_id, rejected_by_id, rejection_reason
WaiverRevoked	training_waiver	waiver_id, revoked_by_id, reason
TrainingMatrixUpdated	training_matrix	changed_cells_json, diff_preview_json, updated_by_id, qa_esignature_id, effective_at
TrainingMatrixApproved	training_matrix	version_number, approved_by_id, esignature_id, effective_at · §11.50
CertificateRevoked	certificate	cert_id, revoked_by_id, reason, esignature_id, user_notified · §11.50
CertificateMarkedObsolete	certificate	cert_id, obsoleted_by_id, reason (actor:SYSTEM for AUTO-03)
CertificateIntegrityChecked	certificate	cert_id, checked_by_id, result:verified violation
CertificateReportExported	report_export	exported_by_id, filter_params, record_count, file_hash · §11.10(b)
SOPDocumentUploaded	document_version	document_id, version, file_hash, is_major_version, uploaded_by_id, change_summary
SOPDocumentPublished	document_version	document_id, version, effective_date, published_by_id, affected_role_ids
RetrainingTriggeredFromSOP	enrollment (batch)	sop_document_id, new_version, affected_count, certs_obsoleted (actor=SYSTEM)
TrainingImpactAssessmentCreated	training_impact_assessment	document_version_id, affected_count, obsoleted_certs, assignments_created
CapaCreated	capa	capa_id, quality_event_id, training_required, created_by_id
CapaTrainingLinked	capa	capa_id, training_assignment_id, linked_by_id

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
CapaMovedToEffectivenessCheck	capa	capa_id, all_training_completed_at (actor=SYSTEM)
CapaClosed	capa	capa_id, closed_by_id, effectiveness_evidence, esignature_id · §11.50
InspectorSessionProvisioned	auditor_session	session_id, access_type, auditor_names, scope, expires_at, provisioned_by_id
InspectorSessionRevoked	auditor_session	session_id, revoked_by_id, revoked_at, reason
InspectionPackageGenerated	inspection_package	package_id, scope, record_count, file_hash, is_official, esignature_id, generated_by_id
AuditIntegrityCheckRun	audit_integrity_result	records_checked, hash_mismatches, sequence_gaps, result (actor=SYSTEM)
SystemConfigChanged	system_configuration	key, old_value, new_value, changed_by_id, esignature_id (security keys)
FeatureFlagToggled	feature_flag	key, old_enabled, new_enabled, toggled_by_id, esignature_id (if required)
ComplianceReportExported	report_export	exported_by_id, scope, filter_params, record_count, file_hash · §11.10(b)
EscalationDispatched	notification	target_user_id, dispatched_by_id, overdue_count, notification_type:escalation
AuditLogExported	report_export	exported_by_id, filter_params, record_count, file_hash, exported_at · §11.10(b)
ScheduledJobManuallyTriggered	scheduled_job_log	job_name, triggered_by_id, triggered_at, reason

18. Admin Portal Pre-Launch QA Checklist

Every item must pass before any admin portal screen ships to production. Run this checklist screen by screen.

18.1 User & org management (10 checks)

1. HRIS-managed fields (name, email, dept, role, hire date) are read-only with "Managed by HRIS" badge in employee profile
2. Employee termination revokes all sessions within 60 seconds (tested with simulated HRIS termination event)
3. Employee termination audit trail: EmployeeTerminated + SessionsRevoked (2 separate rows) both present
4. Bulk import: preview phase shows exact CREATE/UPDATE/ERROR counts before execution
5. Role assignment requires reason field. AuditTrail: UserRoleAssigned includes reason.
6. Training matrix changes show diff preview before QA e-signature prompt
7. Training matrix QA e-signature captured (21 CFR §11.50) with HMAC-SHA256 integrity hash
8. Separation of duties: trainer cannot approve their own courses (UI shows "Cannot approve your own submission")
9. Duplicate assignment detection: warning shown if employee already enrolled in same course version
10. Org hierarchy tree loads within 3s for org with 20,000 employees

18.2 Compliance & certificate management (10 checks)

11. Certificate revocation: QA e-signature required. Persistent red banner appears on employee dashboard immediately after.
12. SOP publication triggers retraining within 15 minutes for affected employees (AUTO-03 tested)
13. CRITICAL SOP retraining deadline = 14 days. Standard SOP = 30 days. Verified against due_date on created assignments.
14. TrainingImpactAssessment row created on SOP publish with correct affected_employee_count.
15. Waiver approval: QA e-signature required. Approved waiver marks employee COMPLIANT in compliance dashboard.
16. Compliance dashboard loads org-wide data for 20k employees within 3s (read replica used for analytics query).
17. Certificate integrity check: "Verify hash" re-computes HMAC server-side. Mismatch dispatches QA alert.
18. Inspection package generated with SHA-256 hash. isOfficial flag requires QA e-signature.
19. Auditor session scoped correctly: auditor cannot read records outside their scope_sites and scope_depts.
20. CAPA closure requires QA e-signature. AuditTrail: CapaClosed with esignature_id.

18.3 System & infrastructure (8 checks)

21. HRIS sync status badge shows last sync time in admin header on every page (not just HRIS monitor screen).

22. Inspector session active badge shows in admin header with countdown timer when auditor session is open.
23. Dead-letter queue alert: red badge appears in admin notification bell within 2 min of DLQ depth > 0.
24. Audit integrity check: nightly job result visible in ADM-13. Manual trigger available. Hash mismatch = immediate alert.
25. Security config changes (session timeout, MFA policy, password policy) require QA e-signature before taking effect.
26. Scheduled job monitor shows last 30 runs for each job. Failed job dispatches admin notification within 5 min.
27. All exports include SHA-256 hash in footer and AuditTrail: ReportExported entry (separate from AccessLog).
28. Admin cannot access employee training content (course builder, question bank) — role guard tested at API level not just UI.

18.4 Visual quality (6 checks)

29. HRIS-managed field indicator ("Managed by HRIS" badge) present on all read-only HRIS fields in employee profile
30. Bulk operation confirmation dialogs show exact affected count ("1,247 employees") before execute
31. Status badges use correct color: ACTIVE (green) | SUSPENDED (amber) | TERMINATED (red) | REVOKED (red bold)
32. Compliance rate rings color-coded correctly: >=90% emerald, 70-89% amber, <70% red
33. All tables have skeleton loaders matching column layout (not generic spinners)
34. CRITICAL SOP badge is orange (#C2410C), clearly distinct from red error states

A P P E N D I C E S

Appendix A — Admin Portal Screen Map

SCREEN	ROUTE	DESCRIPTION
ADM-01	/admin	Admin Dashboard 360° · Phase 1 Critical · <3s load target
ADM-02	/admin/users	User Management · Phase 1 Critical
ADM-03	/admin/users/import + /admin/integrations/hris	Bulk Import & HRIS Monitor · Phase 1 Critical
ADM-04	/admin/roles	Role & Permission Management · Phase 1 Critical
ADM-05	/admin/org	Organisation, Site & Department Management · Phase 1 High
ADM-06	/admin/assignments	Training Assignment Management · Phase 1 Critical
ADM-07	/admin/waivers	Training Waiver Management · Phase 1 High
ADM-08	/admin/training-matrix	Training Matrix Manager · Phase 1 Critical
ADM-09	/admin/certificates	Certificate Management & Revocation · Phase 1 High
ADM-10	/admin/compliance	Compliance Dashboard Org-Wide · Phase 1 Critical
ADM-11	/admin/sop-documents	SOP Document Management · Phase 1 Critical
ADM-12	/admin/capa	CAPA Management · Phase 1 High
ADM-13	/admin/inspection	Inspection Readiness & Audit Package Generator · Phase 1 Critical
ADM-14	/admin/config	System Configuration · Phase 1 High
ADM-15	/admin/reports	Analytics & Reporting Hub · Phase 1 High
ADM-16	/admin/audit-log	Full System Audit Log Viewer · Phase 2 High
ADM-17	/admin/notifications	Notification Management · Phase 2 Medium
ADM-18	/admin/jobs	Scheduled Jobs Monitor · Phase 2 Medium

Appendix B — Admin Role Permission Matrix

ACTION	WHO CAN PERFORM	WHO CAN INITIATE
Create / edit employees	Training Admin, System Admin	QA Officer (read-only)
Terminate employees	Training Admin, System Admin	N/A
Assign training (individual)	Training Admin, Trainer (own dept)	N/A
Assign training (bulk / org-wide)	Training Admin ONLY	N/A
Approve training waivers	QA Officer, QA Director ONLY	Training Admin (request only)
Revoke certificates	QA Officer, QA Director ONLY	Training Admin (flag for review)
Publish SOP documents	QA Officer, Training Admin	Trainer (upload draft only)
Update training matrix	Training Admin (QA e-sig required)	QA Officer (e-sign only)
Provision auditor session	Training Admin, Compliance Officer	QA Officer (view status only)
Generate inspection package	Compliance Officer, Training Admin	QA Officer (e-sign official pkg)
View full audit log	Compliance Officer, System Admin	Training Admin (own org scope)
Change security config	System Admin (QA e-sig required)	Compliance Officer (e-sign only)
Toggle feature flags	System Admin	QA Officer (e-sign for audit-affecting flags)
Manage roles & permissions	System Admin ONLY	N/A
Close CAPA	Compliance Officer, QA Officer (e-sig)	Training Admin (view only)

Appendix C — Key Workflow State Machines

Training Waiver lifecycle

TRANSITION	RULES
PENDING → APPROVED	Trigger: QA Officer completes review + e-signs. employee.compliance = COMPLIANT for this course.
PENDING → REJECTED	Trigger: QA Officer rejects with mandatory reason. Employee notified. No compliance credit.
APPROVED → REVOKED	Trigger: Admin/QA revokes (e.g. new SOP version). Employee must retrain. New assignment created.
APPROVED → EXPIRED	Trigger: waiver.expires_at < NOW(). Automated by daily job. Employee status reverts to non-compliant.

CAPA training requirement lifecycle

TRANSITION	RULES
CAPA CREATED → RCA	Admin completes root cause analysis. rcaCompletedAt set.
RCA → TRAINING REQUIRED	Admin links training assignment. capa.trainingRequired=true.
TRAINING REQUIRED → EFFECTIVENESS CHECK	AUTO-08: All linked employees complete training. System moves CAPA automatically.
EFFECTIVENESS CHECK → CLOSED	Compliance Officer documents effectiveness evidence + QA e-sign.

Appendix D — Admin Analytics Metrics Reference

METRIC	FORMULA + USE CASE
OrgComplianceRate	AVG(individual_compliance_rate) across all active pharma_user rows for this org
DeptComplianceRate	AVG(compliance_rate) per department. Source: department_compliance_snapshot.
OverdueRate	COUNT(enrollment.status=in_progress AND due_date < NOW()) / COUNT(all active enrollments)
RetrainingVelocity	AVG(retraining_enrollment.completed_at - certificate.obsoleted_at) per SOP update event
WaiverRate	COUNT(training_waiver.status=approved) / COUNT(total training assignments) per period
FirstPassRate	COUNT(passed AND attempt_number=1) / COUNT(all first attempts) org-wide
HrisSyncReliability	COUNT(hris_sync_log.status=success) / COUNT(total sync attempts) last 30 days

METRIC	FORMULA + USE CASE
TerminationRevocationLatency	AVG(user_session.ended_at - termination_event.created_at) for termination events. Target: <60s
InspectionReadinessScore	Composite: compliance_rate + audit_trail_integrity + open_findings + cert_integrity. 0-100 score.
CapaTrainingVelocity	AVG(capa.rcaCompletedAt - capa.createdAt) in days + AVG(training completion days after link)

19. ADM-19 — Enrollment Management

Route: /admin/enrollments · Priority: Phase 1 Critical · Regulatory: 21 CFR 11.10(e), ALCOA+

19.1 US-ADM-ENR-001 — Create individual enrollment

"As a Training Administrator, I want to create a single enrollment for a specific employee in a specific course, so that I can target individual training needs with precision."

FIELD	SPECIFICATION
Employee	Search by name or employee ID. Auto-checks for duplicate active enrollment.
Course / Version	Search by title or code. Defaults to latest published version.
Due date	Date picker. Warning shown if < 7 days from today.
Priority	HIGH (red) / MEDIUM (amber, default) / LOW (gray).
Reason / note	Max 500 chars. Shown to employee in their training list.
Mandatory flag	Toggle. Mandatory = counts against compliance rate.

- Duplicate guard: "This employee already has an active enrollment for this course. Proceed to create a second enrollment?"
- AuditTrail: ENROLLMENT_CREATED — user_id, course_version_id, due_date, priority, reason, created_by_id

19.2 US-ADM-ENR-002 — Bulk enrollment via CSV import

"As a Training Administrator, I want to bulk-enroll employees via CSV, so that large cohorts can be assigned training without manual entry."

- CSV template: employee_id, course_code, course_version, due_date, priority, reason
- 3-phase flow: Upload → Preview (new: N, update: U, conflict: C, error: E) → Confirm
- Conflict resolution: if enrollment already exists for same employee+course → skip or UPDATE due date (admin choice per row)
- Error table: row number + column + error message. Partial import allowed.
- Import log: full CSV downloadable. Retained 7 years for audit.
- AuditTrail: BULK_ENROLLMENT_IMPORTED — filename, file_hash, row_count, created_count, updated_count, error_count

19.3 US-ADM-ENR-003 — Modify enrollment (due date, version migration)

"As a Training Administrator, I want to modify an existing enrollment so I can extend due dates or migrate employees to an updated course version."

- Modifiable fields: due_date (extend only, not shorten without reason), course_version (migrate to newer version)
- Version migration: if enrolling in v2.0 when employee has in-progress v1.0 — admin chooses: (a) cancel v1.0 and start fresh, (b) allow v1.0 to complete first
- Due date extension: requires extension_reason. ALCOA+ accountability.
- AuditTrail: ENROLLMENT_MODIFIED — enrollment_id, changed_fields[], old_value_json, new_value_json, modified_by_id, reason

19.4 US-ADM-ENR-004 — Cancel enrollment

"As a Training Administrator, I want to cancel an enrollment with a mandatory reason so that the cancellation is documented and auditable."

- Cancel requires mandatory reason (dropdown + free text)
- Status: enrollment.status = CANCELLED. Not deleted.
- Cannot cancel a COMPLETED enrollment — must use certificate revocation instead
- AuditTrail: ENROLLMENT_CANCELLED — enrollment_id, user_id, course_version_id, cancellation_reason, cancelled_by_id

19.5 US-ADM-ENR-005 — Grant re-enrollment after failed attempts

"As a Training Administrator, I want to reset attempt limits for a failed employee so that they can retake the assessment after receiving additional coaching."

- Re-enrollment: creates a new enrollment_record (old FAILED record immutable)
- Requires justification + approval by QA Officer if course is regulatory-mandatory
- AuditTrail: RE_ENROLLMENT_GRANTED — original_enrollment_id, new_enrollment_id, reason, granted_by_id, approved_by_id

19.6 US-ADM-ENR-006 — Auto-enrollment rules

"As a Training Administrator, I want to configure trigger-based auto-enrollment rules so that training is assigned automatically when conditions are met."

FIELD	SPECIFICATION
Trigger types	employee.hire_date set (new hire) department change job role change certificate expiry approaching SOP update CAPA raised
Rule config	trigger_type + course + due_offset_days + priority + mandatory flag
Rule preview	Shows: "This rule would have triggered 847 times in the last 30 days"
Conflict resolution	If employee already enrolled: skip or extend due date (configurable per rule)
Rule history	All changes to rules version-controlled with change_reason
AuditTrail: AUTO_ENROLLMENT_TRIGGERED — rule_id, user_id, trigger_type, enrollments_created (actor=SYSTEM)	

19.7 US-ADM-ENR-007 — Employee learning transcript

"As a Training Administrator, I want to view and export the complete training history for any employee as a formal learning transcript."

- Route: /admin/users/:userId/transcript
- Transcript shows: all enrollments (completed/cancelled/failed), certificates, assessment attempts, e-signatures, material_progress records
- Export: PDF with digital signature block + SHA-256 hash + watermark "Official Training Transcript"
- ALCOA+ footer: timestamps in UTC, version trained on, e-signature IDs
- AuditTrail: TRANSCRIPT_EXPORTED — target_user_id, exported_by_id, record_count, file_hash

19.8 Events & AuditTrail — Enrollment Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
ENROLLMENT_CREATED	enrollment_records	user_id, course_version_id, due_date, priority, mandatory, reason, created_by_id
BULK_ENROLLMENT_IMPORTED	import_jobs	filename, file_hash, row_count, created_count, updated_count, error_count, imported_by_id
ENROLLMENT_MODIFIED	enrollment_records	enrollment_id, changed_fields[], old_value_json, new_value_json, reason, modified_by_id
ENROLLMENT_CANCELLED	enrollment_records	enrollment_id, user_id, course_version_id, cancellation_reason, cancelled_by_id
RE_ENROLLMENT_GRANTED	enrollment_records	original_enrollment_id, new_enrollment_id, reason, granted_by_id
AUTO_ENROLLMENT_TRIGGERED	enrollment_rules	rule_id, user_id, trigger_type, enrollments_created (actor=SYSTEM)
TRANSCRIPT_EXPORTED	report_export	target_user_id, exported_by_id, record_count, file_hash · §11.10(b)

20. ADM-20 — Batch & Cohort Training Management

Route: /admin/batches · Priority: Phase 1 High · Regulatory: GMP Annex 11 §4

"As a Training Administrator, I want to group employees into training batches and manage the entire cohort lifecycle — from creation through completion reporting — so that large-scale training rollouts are coordinated and auditable."

20.1 US-ADM-BAT-001 — Create training batch

FIELD	SPECIFICATION
Batch name	Unique name. E.g. "GMP Fundamentals — Site A — Q2 2026"
Training programme	One or more courses in order. Each with due_date offset from batch start.
Start date	Date all enrollments will be created. Can be future-dated.
Target audience	Department / Job role / Site / Manual employee list
Status	DRAFT (editable) → ACTIVE (enrollments created) → COMPLETED / CLOSED
Prerequisites check	System validates: all batch members meet course prerequisites. Flags exceptions.

- AuditTrail: BATCH_CREATED — batch_id, name, course_count, target_audience, start_date, created_by_id

20.2 US-ADM-BAT-002 — Manage batch membership

- Add members: search + multi-select employees. Preview count.
- Remove members: only allowed if batch is DRAFT or if employee has not yet started any batch course
- Bulk add from HRIS query: filter by department/job role/site/hire date range
- Membership roster: downloadable CSV with employee_id, name, department, site
- AuditTrail: BATCH_MEMBER_ADDED / BATCH_MEMBER_REMOVED — batch_id, user_id, changed_by_id, reason (for removal)

20.3 US-ADM-BAT-003 — Activate batch and create enrollments

- "Activate Batch" button: transitions DRAFT → ACTIVE
- Pre-activation summary: "This will create N enrollments for M employees across P courses"
- On activation: bulk INSERT enrollment_records for all batch_members x batch_courses
- Notifications dispatched to all batch members simultaneously

- AuditTrail: BATCH_ACTIVATED — batch_id, enrollments_created, notifications_sent, activated_by_id

20.4 US-ADM-BAT-004 — Monitor batch progress

- Progress dashboard per batch: completion % ring, per-course progress bars, employee-level status table

METRIC	DISPLAY
Overall progress	COUNT(completed) / COUNT(total enrollments) across all courses
Per-course breakdown	Bar chart: not started / in progress / completed / overdue / failed per course
Per-employee status	Table: employee + courses completed + courses overdue + courses in progress
Overdue alerts	Red badge on batch card when any member is > 3 days overdue
Send bulk reminder	Targets only batch members who have not started or are overdue

20.5 US-ADM-BAT-005 — Modify active batch

- Add course to active batch: new enrollments created only for members who have not completed it
- Extend batch deadline: extends due_date on all uncompleted enrollments in batch
- Extension requires reason. AuditTrail: BATCH_MODIFIED

20.6 US-ADM-BAT-006 — Generate batch compliance report

- Report: all batch members, per-course completion status, scores, certificates, e-signature IDs
- Export: PDF (digitally signed) or CSV. SHA-256 hash in footer.
- AuditTrail: BATCH_REPORT_GENERATED — batch_id, exported_by_id, record_count, file_hash

20.7 Events & AuditTrail — Batch Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
BATCH_CREATED	training_batches	batch_id, name, course_count, audience_type, start_date, created_by_id
BATCH_ACTIVATED	training_batches	batch_id, enrollments_created, notifications_sent, activated_by_id
BATCH_MEMBER_ADDED	batch_members	batch_id, user_id, added_by_id
BATCH_MEMBER_REMOVED	batch_members	batch_id, user_id, removed_by_id, reason
BATCH_MODIFIED	training_batches	batch_id, changed_fields[], old/new values, reason, modified_by_id
BATCH_REPORT_GENERATED	report_export	batch_id, exported_by_id, record_count, file_hash · Annex 11 §4

21. ADM-21 — Assessment Management (Admin View)

Route: /admin/assessments · Priority: Phase 1 High · Regulatory: 21 CFR 11.10(e), ALCOA+

21.1 US-ADM-ASS-001 — Question bank management

"As a Content Administrator, I want to manage a global question bank with versioned, approved questions so that all assessments draw from controlled, validated question pools."

- Question lifecycle: DRAFT → REVIEWED → APPROVED → ACTIVE (same approval workflow as courses)
- Question types: single choice, multiple choice, true/false, short text, scenario-based
- Fields per question: text, type, category, difficulty, explanation, regulation_reference (e.g. 21 CFR 211.68)
- Versioning: editing a question used in a published assessment creates a new version; old version immutable
- Bulk import: CSV or XML (QTI format). Validation report before commit.
- AuditTrail: QUESTION_APPROVED — question_id, version, approved_by_id, approved_at

21.2 US-ADM-ASS-003 — Review assessment attempt records

"As a Training Administrator, I want to view individual assessment attempt records so I can answer disputes or audit questions about exactly what was submitted."

- Attempt detail view: start time, submit time, duration, IP address, device fingerprint, score
- Per-question view: question text + employee's selected answer + correct answer + points awarded
- All attempt data is read-only — no edit or delete affordance
- Admin annotation: append notes to attempt (INSERT attempt_annotations, never modifies original)
- AuditTrail: ASSESSMENT_ATTEMPT_VIEWED — attempt_id, viewed_by_id, viewed_at

21.3 US-ADM-ASS-004 — Override assessment result (two-person rule)

"As a Training Administrator, I want to override an assessment result in exceptional circumstances so a legitimate learner is not penalised by a system error, with full two-person accountability."

Override compliance requirements (21 CFR Part 11)
Override requires SUPER_ADMIN or QA_REVIEWER level authorisation

Two-person rule enforced: one admin requests, a different admin approves
Original attempt record is NEVER modified — override creates a separate assessment_overrides record
Override record: original_attempt_id, override_score, override_reason, requested_by_id, approved_by_id, signed_at
If override score meets passing threshold: certificate is issued automatically
AuditTrail: ASSESSMENT_OVERRIDE_APPLIED with BOTH signatories in key fields

21.4 US-ADM-ASS-005 — Minimum time enforcement configuration

- min_time_seconds configurable per assessment by admin
- Server-side enforcement: submission rejected if (submitted_at – started_at) < min_time_seconds
- Client-side timer shows remaining minimum time before Submit button activates
- Bypass attempt via API: UPDATE assessment_attempt SET status=INVALID, violation_type=MIN_TIME
- AuditTrail: ASSESSMENT_MIN_TIME_VIOLATION — attempt_id, time_elapsed_seconds, min_required_seconds

21.5 US-ADM-ASS-006 — Proctoring configuration

PROCTORING LEVEL	SPECIFICATION
NONE	No proctoring. Standard anti-cheating (randomise only).
HONOUR	Declaration only: employee ticks "I confirm I am completing this assessment without assistance."
MONITORED	Webcam capture. Stored in S3 for configurable N days. Admin can review.
STRICT	Webcam + screen recording. Browser lockdown (prevent copy-paste, tab switching, external links). Full proctoring.

- MONITORED/STRICT: proctoring_events table logs tab-switch, camera-blocked, copy-paste violations
- Admin review: flagged attempts shown with violation markers and footage timestamp

21.6 US-ADM-ASS-007 — Assessment analytics report

- Metrics: pass rate (overall + by department), average score, score distribution histogram, per-question difficulty index
- Question difficulty: (wrong_count / total_attempts) per question_id. Questions with difficulty > 70% wrong flagged in red.
- Trend: pass rate over 6-month rolling window per course version
- Filter: course, department, date range
- Export: CSV + PDF. AuditTrail: ASSESSMENT_ANALYTICS_EXPORTED

21.7 Events & AuditTrail — Assessment Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
QUESTION_APPROVED	assessment_questions	question_id, version, approved_by_id, approved_at
ASSESSMENT_ATTEMPT_VIEWED	assessment_attempts	attempt_id, viewed_by_id, viewed_at
ASSESSMENT_OVERRIDE_APPLIED	assessment_overrides	override_id, original_attempt_id, override_score, reason, requested_by_id, approved_by_id · §11.10(e)
ASSESSMENT_MIN_TIME_VIOLATION	assessment_attempts	attempt_id, time_elapsed_seconds, min_required_seconds (actor=SYSTEM)
ASSESSMENT_ANALYTICS_EXPORTED	report_export	course_version_id, filter_params, exported_by_id, file_hash

22. ADM-22 — Notifications & Communications Management

Route: /admin/notifications · Priority: Phase 2 High · Regulatory: GxP data integrity

22.1 US-ADM-NOT-001 — Notification template management

"As a Training Administrator, I want to configure notification templates for all system events so communications reflect our brand and contain the right information."

FIELD	SPECIFICATION
Template types	enrollment_created, enrollment_overdue, assessment_reminder, certificate_issued, sop_assigned, waiver_approved, capa_training_required, system_maintenance
Template fields	Subject line, HTML body (rich text), priority (mandatory/high/normal/low)
Variables	{{employee_name}}, {{course_name}}, {{due_date}}, {{portal_link}}, {{days_overdue}}, {{manager_name}}
Multi-language	Each template can have locale variants. System selects based on user.preferred_language.
Test send	Admin enters recipient email → sends real test notification with sample variable values
AuditTrail: NOTIFICATION_TEMPLATE_UPDATED — template_type, changed_by_id, old_version_archived	

22.2 US-ADM-NOT-002 — Automated reminder rules

"As a Training Administrator, I want to configure when and how often employees are reminded about pending training so compliance rates improve without manual effort."

RULE FIELD	SPECIFICATION
Trigger types	days_before_due (e.g. -30, -14, -7, -3) days_after_due (overdue escalation)
Frequency	once daily weekly until completed
Channel	email in-app both

RULE FIELD	SPECIFICATION
Quiet hours	Do not send between configurable hours (default 22:00–08:00 local time)
Escalation rule	If employee ignores after N reminders → notify direct manager - Reminder count tracked: enrollment_records.reminder_count + last_reminder_at - AuditTrail: REMINDER_RULE_UPDATED — rule_id, changed_by_id, old/new config

22.3 US-ADM-NOT-003 — Broadcast announcements

"As a Training Administrator, I want to send broadcast announcements to targeted groups so I can communicate urgent compliance updates without email spam."

- Recipient scope: All / Department / Site / Job Spec / Individual users
- Channel: email, in-app, or both
- Scheduled send: future date and time. Preview recipient count before sending.
- Read tracking: which recipients have opened/dismissed the in-app notification
- AuditTrail: ANNOUNCEMENT_SENT — announcement_id, recipient_count, scope, sent_by_id

22.4 US-ADM-NOT-004 — Escalation chain configuration

"As a Training Administrator, I want to configure escalation chains for overdue training so the system automatically notifies up the management chain."

ESCALATION LEVEL	RECIPIENTS
Day 1 overdue	Employee only
Day 3 overdue	Employee + direct manager
Day 7 overdue	Employee + direct manager + department head
Day 14 overdue	All above + site QA officer
Escalation stop	Stops automatically when enrollment is COMPLETED or CANCELLED - Thresholds configurable per rule. Default values shown above. - Escalation events logged: escalation_events table (enrollment_id, level, sent_to, sent_at)

22.5 US-ADM-NOT-005 — Notification delivery logs

"As a Training Administrator, I want to view notification delivery logs and open rates so I can verify reminders were sent and identify delivery failures."

- Log columns: notification type, recipient, channel, sent_at, status (sent/failed/bounced), opened (email pixel tracking, configurable)
- Filter: type, recipient, date range, status
- "Resend failed" button: re-publishes to Kafka. AuditTrail: NOTIFICATION_RESENT
- Delivery failure alert: red bell badge when notification_log has status=failed in last 24h

22.6 US-ADM-NOT-006 — User notification preferences

- Admin can configure default preferences per notification type for all users
- User-level: channel (email/in-app), frequency (immediate/daily digest/weekly)
- Mandatory notification types cannot be opted out by employees (21 CFR compliance events: overdue, cert revoked, SOP retraining)
- Admin override: force a user’s preferences to a specific channel (e.g. for users without email access)

22.7 Events & AuditTrail — Notifications Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
NOTIFICATION_TEMPLATE_UPDATED	notification_templates	template_type, changed_by_id, old_version_id
ANNOUNCEMENT_SENT	announcements	announcement_id, recipient_count, scope, channel, sent_by_id
NOTIFICATION_RESENT	notification_logs	original_notification_id, resent_by_id, resent_at

23. ADM-23 — Analytics & Business Intelligence

Route: /admin/analytics · Priority: Phase 1 High · Regulatory: GMP Annex 11 §9

23.1 US-ADM-ANL-001 — Training analytics dashboard

"As a Training Administrator, I want a training analytics dashboard showing learner engagement and content effectiveness so I can identify which courses need improvement."

METRIC	DESCRIPTION
Avg time-to-completion	Per course. Histogram showing distribution. Outliers flagged (too fast = gaming; too slow = confusion).
Avg score + trend	Per course version. 6-month trend line. Version comparison toggle.
Completion rate	COUNT(completed) / COUNT(enrolled) per course. Sorted worst-first.
Failure rate + re-attempt rate	Courses with failure > 20% flagged for content review.
Learner engagement	Avg sessions per enrollment, dropout rate (sessions with no completion in 30d).
Content effectiveness	Pass rate trend after each content update — did the update improve outcomes?

- Drill down: org → department → individual learner
- Materialized views refreshed nightly. On-demand refresh button for urgent reporting.

23.2 US-ADM-ANL-002 — Custom report builder

"As a Training Administrator, I want to build custom reports without developer support so I can answer ad-hoc queries."

- Fields available: all user, enrollment, session, assessment, certificate, audit trail fields
- Operations: filter (WHERE), group by, aggregate (count, avg, sum, min, max)
- Saved reports: name + visibility (private/shared with role)
- Scheduled delivery: saved report auto-generated and emailed (cron expression or frequency)
- Output: table view in browser + export CSV/PDF
- AuditTrail: CUSTOM_REPORT_EXECUTED — report_id, report_name, filter_params, row_count, executed_by_id

23.3 US-ADM-ANL-003 — Predictive compliance forecasting

"As a Training Administrator, I want to see which departments are at risk of falling below 80% compliance in the next 30/60/90 days so I can act proactively."

- Model: based on current due dates, historical completion pace per employee/department, in-progress rates
- Output: department × predicted compliance % at 30d/60d/90d horizon
- Risk flags: departments predicted to drop below configured threshold highlighted in amber/red
- Suggested actions: bulk reminder, deadline extension, resource allocation
- Nightly job: stores in compliance_forecasts {department_id, horizon_days, predicted_pct, calculated_at}

23.4 US-ADM-ANL-004 — Training ROI and cost analysis

- Metrics: total completions per month, avg time-to-complete, estimated cost per completion (content_cost / completions)
- Cost fields: admin enters content_cost per course (content creation cost + licensing). Not synced from finance system.
- Trend: completion volume year-over-year. Time saved vs manual tracking estimate (configurable baseline).

23.5 US-ADM-ANL-005 — Site/department benchmark comparison

- Side-by-side: compliance %, overdue rate, avg completion time, pass rate for selected departments or sites
- Organisation average as benchmark line on all charts
- League table: ranking of departments by compliance % (opt-in feature for site directors; privacy-configurable)
- Export: PDF benchmarking report for QA submission

23.6 Scheduled reports (US-ADM-RPT-004)

"As a Training Administrator, I want to configure scheduled reports that are automatically generated and emailed to stakeholders."

- Configuration: frequency (daily/weekly/monthly), recipient list, report type, filter scope, format (PDF/CSV)
- Recipients: specific users or users with a particular role (e.g. all QA_REVIEWER)
- Failure notification: if scheduled report generation fails, admin alerted within 5 min
- AuditTrail: SCHEDULED_REPORT_SENT — report_id, cron_expression, recipient_count, file_hash

24. ADM-24 — System Configuration & Integrations (Complete)

Route: /admin/system · Priority: Phase 1 High · Regulatory: 21 CFR 11.30, GAMP 5 §10

24.1 US-ADM-SYS-001 — Portal-wide settings & organisation branding

"As a System Administrator, I want to configure org name, logo, timezone, date format, and language defaults."

SETTING	SPECIFICATION
org_name	Organisation legal name. Appears on certificates and reports.
logo_s3_key	Upload logo (SVG/PNG, max 500KB). Appears on all portals and certificate PDFs.
primary_timezone	IANA timezone (e.g. "Asia/Kolkata"). All timestamps displayed in this TZ; stored in UTC.
date_format	Default: MMM DD, YYYY (e.g. "Mar 21, 2026"). Required format for ALCOA+ clarity.
default_language	BCP 47 locale code (e.g. "en-US"). Used when user has no language preference set.
cert_validity_default_months	Default cert expiry for courses that do not specify their own. Default: 12.

- All changes: AuditTrail SYSTEM_CONFIG_UPDATED with old_value and new_value

24.2 US-ADM-USR-005 — SSO / LDAP configuration

"As a System Administrator, I want to configure SSO/LDAP so employees log in with corporate credentials."

- SSO: SAML 2.0 or OIDC. Fields: provider URL, entity ID, certificate upload, attribute mapping (email, firstName, department, jobRole)
- LDAP: server URL, bind DN, base DN, attribute mapping, TLS configuration
- "Test connection" button: verifies connectivity before saving. SAML: performs SP-initiated auth flow in sandbox.
- Fallback: if SSO unavailable, local password auth allowed for ADMIN and AUDITOR roles only
- JIT provisioning: new SSO user auto-creates LMS account with EMPLOYEE role on first login

- Group-to-role mapping: AD group name → LMS role (configurable table)
- AuditTrail: SSO_CONFIG_UPDATED — provider_type, changed_by_id, test_result

24.3 US-ADM-SYS-003 — API key management

"As a System Administrator, I want to manage API keys for external integrations so external systems consume LMS data securely."

- API key fields: name, description, scopes (read:enrollments, read:certificates, write:enrollments, admin:users), expiry date
- Keys stored as bcrypt hashes. Shown once in plaintext on creation. Copy-to-clipboard button.
- Rate limiting: configurable requests/minute per key
- "Revoke" button: immediate effect. AuditTrail: API_KEY_REVOKED
- Usage stats: last used timestamp, request count (30d)
- AuditTrail: API_KEY_CREATED / API_KEY_REVOKED — key_name, scopes, created_by_id/revoked_by_id, revoked_reason

24.4 US-ADM-SYS-004 — Kafka message bus & DLQ management

"As a System Administrator, I want to manage Kafka topics and dead-letter queues so asynchronous events are reliably processed."

- Kafka health panel: consumer lag per topic, messages-per-second, error rate
- Topics list: retention days, partition count, replication factor. Admin can edit retention.
- DLQ panel: dead_letter_queue rows with failure_reason, retry_count, failed_event_payload
- "Retry" button: re-publishes event to original topic. AuditTrail: DLQ_EVENT_RETRIED
- "Discard" button: marks event as manually resolved. AuditTrail: DLQ_EVENT_DISCARDED — requires reason
- Alert: red badge in admin header when DLQ depth > 0

24.5 US-ADM-SYS-005 — System health dashboard

"As a System Administrator, I want a system health dashboard with infrastructure metrics so I can detect issues before they impact learners."

METRIC	SPECIFICATION
DB connection pool	Current used / max. Colour-coded. Alert at > 80%.
Query latency P95	Milliseconds per endpoint. 30-day trend. Alert at > 1000ms.
S3 storage used	GB used. Alert at > 80% of provisioned.
Kafka consumer lag	Per topic. Alert at > 10,000 messages unprocessed.

METRIC	SPECIFICATION
API error rate	% of requests returning 4xx/5xx. Alert at > 1%.
Session count	Active user sessions. Spike detection (> 2x normal → alert).

- Historical: 30-day trend for all metrics. Alerts: in-app notification + email to System Admin role.

24.6 US-ADM-SYS-006 — GAMP 5 validation document vault

"As a System Administrator, I want to manage the validation documentation required for GAMP 5 / 21 CFR Part 11 computerised system validation."

- Documents managed: URS (User Requirements Spec), FS (Functional Spec), DQ (Design Qualification), IQ, OQ, PQ, Validation Summary Report
- Each document: type, version, status (DRAFT/APPROVED), approved_by, approval_date, linked_software_version, s3_key, file_hash
- Immutable once APPROVED: no overwrite. New version creates new record.
- Re-validation trigger: on software version change, system flags validation docs for review (needs_review=true)
- AuditTrail: VALIDATION_DOCUMENT_UPLOADED / VALIDATION_DOCUMENT_APPROVED — doc_type, version, approved_by_id

24.7 US-ADM-SYS-007 — Backup & disaster recovery

"As a System Administrator, I want to configure backup and DR settings and test restore procedures to meet our RTO/RPO targets."

SETTING	SPECIFICATION
Backup schedule	Daily full + hourly incremental. Stored to S3 cross-region (different AWS region).
RPO target	≤ 1 hour. Hourly incremental ensures max 1 hour of data loss.
RTO target	≤ 4 hours. Tested quarterly via restore drill to staging environment.
Manual backup	Trigger on-demand before major changes. AuditTrail: MANUAL_BACKUP_TRIGGERED.
Restore test	Quarterly restore drill: restore to staging, run integrity check, document result.
Encryption	All backups encrypted at rest (AES-256) and in transit (TLS 1.3). Key rotation configurable.

- All backup/restore operations logged: backup_logs table with file_hash, size_bytes, duration_seconds
- AuditTrail: BACKUP_COMPLETED / RESTORE_TEST_COMPLETED — s3_key, file_hash, size_bytes, duration_seconds, result

24.8 Events & AuditTrail — System Config Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
SSO_CONFIG_UPDATED	system_configuration	provider_type, changed_by_id, test_result · §11.300
API_KEY_CREATED	api_keys	key_name, scopes, expiry, created_by_id · §11.30
API_KEY_REVOKED	api_keys	key_name, revoked_by_id, revoked_at, reason · §11.30
DLQ_EVENT_RETRIED	dead_letter_queue	event_id, topic, retried_by_id, retried_at
BACKUP_COMPLETED	scheduled_job_log	s3_key, file_hash, size_bytes, result (actor=SYSTEM) · §11.10(c)
VALIDATION_DOCUMENT_APPROVED	system_configuration	doc_type, version, approved_by_id · GAMP 5

25. ADM-25 — Data Governance, Archival & GDPR

Route: /admin/data-governance · Priority: Phase 2 High · Regulatory: 21 CFR 11.10(c), EU Annex 11 §17, GDPR Art.17/20

25.1 US-ADM-DGV-001 — Data retention policies

"As a Data Governance Administrator, I want to configure and enforce data retention policies per record type so we retain records for the minimum regulatory period while managing storage costs."

RECORD TYPE	DEFAULT RETENTION PERIOD
training_record / certificate	7 years from completion date (21 CFR 211.68 standard)
audit_trail	10 years (EU GMP Annex 11 requirement for regulated systems)
assessment_attempts	7 years
material_progress (engagement logs)	5 years
proctoring video	2 years (configurable)
notification_logs	3 years
access_log	5 years

- Records approaching retention end: flagged 90 days in advance for admin review
- Archive: S3 lifecycle rule moves to Glacier (cold tier). Still queryable but slower. Never deleted.
- System prevents deletion of records still within retention period — "Delete" button disabled with tooltip explanation
- Archive confirmation requires e-signature (QA Officer). AuditTrail: RECORDS_ARCHIVED

25.2 US-ADM-DGV-002 — GDPR right-to-erasure (with regulatory carve-out)

"As a Data Governance Administrator, I want to handle GDPR erasure requests while preserving all regulatory-required records — erasing personal contact data and pseudonymising names in GxP records."

GDPR erasure vs GxP retention conflict resolution

GxP records (audit_trail, training_sessions, certificates, assessment_attempts) are NEVER deleted — retained under 21 CFR
Personal contact data (email, phone, home address) CAN be erased — not required for GxP compliance
Pseudonymisation: first_name, last_name → PSEUDONYMISED-{UUID}; email → anonymised-{uuid}@anon.local
After pseudonymisation, training records remain queryable by employee_id but name is anonymised
Process requires legal review before execution
AuditTrail: GDPR_ERASURE_EXECUTED — user_id_hash (NOT the actual ID), fields_pseudonymised_count

- DSAR workflow: admin receives request → INSERT data_subject_requests {type=ERASURE} → legal review step → approved → execute
- 30-day SLA timer visible on pending DSAR requests

25.3 US-ADM-DGV-003 — Data portability export for departing employees

"As a Data Governance Administrator, I want to export a complete data package for a departing employee in machine-readable format (GDPR Art. 20)."

- Export content: all enrollment_records, certificates, assessment_attempts, audit trail events, material_progress for the user
- Format: JSON (machine-readable) + PDF human-readable summary
- Security: password-protected ZIP — password set by employee or delivered via separate channel. NOT email attachment.
- Secure link: S3 pre-signed URL with 15-minute expiry. Logged in S3 access log.
- AuditTrail: DATA_EXPORT_GENERATED — target_user_id, exported_by_id, format, record_count, file_hash

25.4 US-ADM-DGV-004 — Data classification management

CLASSIFICATION	RECORDS + RULES
REGULATORY_RECORD	audit_trail, training_sessions, certificates, assessment_attempts. Highest protection. Never delete. Encrypt at rest.
PERSONAL_DATA	user.email, user.first_name, user.last_name, user.phone. Subject to GDPR erasure. Encrypt at rest + in transit.
CONFIDENTIAL	course content, SOP documents, assessment questions. Internal use only.
PUBLIC	certificate QR verification data (holder name + course + status only).

- Data catalog: visual map of which tables contain which classification label

- Auto-classification rules configured in system settings; applied as column tags
- Used by backup encryption, access control, and retention policies

25.5 US-ADM-DGV-005 — Data lineage report

"As a Data Governance Administrator, I want a data lineage report showing how training records flow from creation to archival so I can explain data provenance to inspectors and GDPR auditors."

- Lineage chain: HR system → pharma_user → enrollment_records → training_sessions → assessment_attempts → certificates → audit_trail → archive tier
- Each node: data source, processing service, storage location, encryption type, retention rule, access control policy
- Volume metrics: current row count per table, data size, growth rate
- Export: PDF for compliance officer review. AuditTrail: DATA_LINEAGE_REPORT_GENERATED

25.6 Events & AuditTrail — Data Governance Module

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
RECORDS_ARCHIVED	retention_archive	entity_type, record_count, archive_tier, e_signature_id, archived_by_id · §11.10(c)
GDPR_ERASURE_EXECUTED	pharma_user	user_id_hash (NOT user_id), fields_pseudonymised_count, approved_by_id · GDPR Art.17
DATA_EXPORT_GENERATED	report_export	target_user_id, exported_by_id, record_count, file_hash, export_format · GDPR Art.20
DATA_LINEAGE_REPORT_GENERATED	access_log	generated_by_id, generated_at

26. ADM-26 — Privileged Access Review & Advanced User Management

Completes Module 1 — User & Identity Management user stories not covered in ADM-02 through ADM-04.

26.1 US-ADM-USR-003 — SUPER_ADMIN two-person rule

"As a System Administrator, I want SUPER_ADMIN role assignment to require dual approval so we maintain two-person integrity for the highest privilege level."

- SUPER_ADMIN assignment: first admin requests → second admin (not the same person) approves → role takes effect
- Pending SUPER_ADMIN requests visible in admin notification bell until approved or rejected
- Role history: every role change INSERT into role_history (append-only, never updated)
- role_history fields: user_id, role_id, action (ADDED/REMOVED), changed_by_id, change_reason, approved_by_id, changed_at
- JWT cache invalidated on role change: Redis DEL user:{id}:session
- AuditTrail: USER_ROLE_ASSIGNED — user_id, role_id, assigned_by_id, approved_by_id, reason

26.2 US-ADM-USR-008 — Quarterly privileged access review

"As a Training Administrator, I want to conduct a quarterly access review of all ADMIN/AUDITOR/QA_REVIEWER accounts so we maintain least-privilege compliance for FDA inspections."

- Trigger: scheduled job fires ACCESS_REVIEW_INITIATED on first day of review quarter
- Review screen: table of all users with elevated roles — user, role, date_granted, last_login, granted_by_admin
- Admin must action each record: "Recertify" (confirm access remains appropriate) or "Revoke" (triggers deactivation workflow)
- Review window: configurable days (default 30). Unreviewed records auto-revoke at window close (configurable).
- Review result: stored in access_reviews table with digital signature (21 CFR §11.50)
- Export: signed PDF of review results for regulatory submission. AuditTrail: ACCESS_REVIEW_COMPLETED

26.3 US-ADM-RPT-006 — Regulatory traceability matrix

"As a Training Administrator, I want a full traceability matrix linking regulations to SOPs to training completions so I can answer any FDA 483 question about regulatory coverage."

- Matrix chain: Regulation (e.g. 21 CFR 211.68) → SOP version → Linked course(s) → Employees who completed
- Filter: regulation type, department, site, date range
- Export: multi-sheet Excel with navigation links between sections. Each sheet is one level of the chain.
- Use case: "Show me all employees trained on the procedures required by 21 CFR 211.68 as of today"
- AuditTrail: TRACEABILITY_MATRIX_EXPORTED — filter_params, row_count, exported_by_id, file_hash

26.4 US-ADM-RPT-007 — Compliance health score (weighted 0–100)

"As a Training Administrator, I want a single 0–100 compliance health score so senior management and QA leadership have one KPI for training compliance."

SCORE COMPONENT	FORMULA
Enrollment completion rate (40%)	COUNT(completed) / COUNT(total active enrollments)
Overdue resolution rate (30%)	1 – (overdue_count / total_assignments). Lower overdue = higher score.
Certificate currency (20%)	COUNT(active certs) / (COUNT(active) + COUNT(expired + obsolete)). Measures cert freshness.
Acknowledgement rate (10%)	COUNT(acknowledged SOPs) / COUNT(assigned SOPs requiring acknowledgement).

- Weighting configurable by admin. Above are defaults.
- Score: 0–100. >90 = green, 70–90 = amber, <70 = red.
- Trend: 12-month history. Stored in compliance_scores {score, component_scores JSONB, calculated_at}.

26.5 US-ADM-AUD-002 — HMAC chain integrity verification

"As a Training Administrator, I want to run HMAC chain integrity verification to confirm no audit trail records have been tampered with."

- Algorithm: for each audit_trail record in sequence order, recompute HMAC-SHA256(prev_hash + id + key_fields + secret_key)
- Compare computed hash to stored hmac_hash. Report: PASS (all match) or FAIL (list of records with mismatch).
- Scheduled: monthly automated run. On-demand: admin triggers via button.
- Verification certificate: PDF with admin name, timestamp, records_checked, violations_found, run_duration
- If violations found: INSERT integrity_violations {record_id, expected_hash, stored_hash}. Immediate alert to SUPER_ADMIN.

- AuditTrail: INTEGRITY_CHECK_RUN — records_checked, violations_found, result:pass|fail, run_by_id

26.6 US-ADM-AUD-004 — E-signature integrity review

"As a Training Administrator, I want to review e-signature records and verify their integrity hashes."

- View: all e_signatures filterable by user, action type, date range, entity
- Each record: user, action_meaning, timestamp UTC, integrity_hash (masked), ip_address, user_agent
- "Verify" button: recomputes HMAC-SHA256(signature_data + user_id + timestamp + secret_key). Shows VALID or TAMPERED.
- TAMPERED result: dispatches immediate QA alert + CertificateIntegrityViolation audit entry
- AuditTrail: ESIGNATURE_INTEGRITY_CHECKED — signature_id, result:valid|tampered, checked_by_id

26.7 US-ADM-AUD-005 — Login & session audit per user

"As a Training Administrator, I want to view a full login and session audit for any user so I can reconstruct their activity for a security incident or regulatory inquiry."

- Report: all login events, session durations, IP addresses, devices for a specific user in a date range
- Shows: successful logins, failed attempts, account lockouts, MFA events, session timeout events
- Chronological timeline view with diff-style colour coding (success = green, failure = red)
- Export: PDF with admin signature. AuditTrail: USER_ACTIVITY_REPORT_VIEWED — target_user_id, viewed_by_id

26.8 Events & AuditTrail — Additional stories

AUDITTRAIL ACTION	ENTITY TYPE	KEY FIELDS
ACCESS_REVIEW_COMPLETED	access_reviews	review_id, reviewed_by_id, recertified_count, revoked_count, signed_at · §11.10(g)
USER_REACTIVATED	pharma_user	user_id, reactivated_by_id, reactivated_at, reason
INTEGRITY_CHECK_RUN	audit_integrity_result	records_checked, violations_found, result, run_by_id · §11.10(e)
ESIGNATURE_INTEGRITY_CHECKED	electronic_signature	signature_id, result:valid tampered, checked_by_id · §11.50
TRACEABILITY_MATRIX_EXPORTED	report_export	filter_params, row_count, exported_by_id, file_hash · Annex 11 §9
COMPLIANCE_SCORE_CALCULATED	analytics_snapshot	score, component_scores_json, calculated_at (actor=SYSTEM)
USER_ACTIVITY_REPORT_VIEWED	access_log	target_user_id, viewed_by_id, date_range, viewed_at

U P D A T E D A P P E N D I C E S

Appendix E — Complete Screen Map (v2.0 — 26 Screens)

SCREEN	ROUTE	DESCRIPTION
ADM-01	/admin	Admin Dashboard 360° · Phase 1 Critical
ADM-02	/admin/users	User Management · Phase 1 Critical
ADM-03	/admin/users/import + /admin/integrations/hris	Bulk Import & HRIS Monitor · Phase 1 Critical
ADM-04	/admin/roles	Role & Permission Management · Phase 1 Critical
ADM-05	/admin/org	Organisation, Site & Department · Phase 1 High
ADM-06	/admin/assignments	Training Assignment Management · Phase 1 Critical
ADM-07	/admin/waivers	Training Waiver Management · Phase 1 High
ADM-08	/admin/training-matrix	Training Matrix Manager · Phase 1 Critical
ADM-09	/admin/certificates	Certificate Management & Revocation · Phase 1 High
ADM-10	/admin/compliance	Compliance Dashboard Org-Wide · Phase 1 Critical
ADM-11	/admin/sop-documents	SOP Document Management · Phase 1 Critical
ADM-12	/admin/capa	CAPA Management (ICH Q10) · Phase 1 High
ADM-13	/admin/inspection	Inspection Readiness & Package Generator · Phase 1 Critical
ADM-14	/admin/config	System Security & Compliance Config · Phase 1 High
ADM-15	/admin/reports	Analytics & Reporting Hub · Phase 1 High
ADM-16	/admin/audit-log	Full System Audit Log Viewer · Phase 2 High
ADM-17	/admin/notifications	Notification Management · Phase 2 Medium
ADM-18	/admin/jobs	Scheduled Jobs Monitor · Phase 2 Medium

SCREEN	ROUTE	DESCRIPTION
ADM-19	/admin/enrollments	Enrollment Management (Module 3) · Phase 1 Critical
ADM-20	/admin/batches	Batch & Cohort Management (Module 4) · Phase 1 High
ADM-21	/admin/assessments	Assessment Management Admin View (Module 6) · Phase 1 High
ADM-22	/admin/notifications (full)	Notifications & Communications (Module 11) · Phase 2 High
ADM-23	/admin/analytics	Analytics & Business Intelligence (Module 12) · Phase 1 High
ADM-24	/admin/system	System Configuration & Integrations (Module 13) · Phase 1 High
ADM-25	/admin/data-governance	Data Governance, Archival & GDPR (Module 14) · Phase 2 High
ADM-26	/admin/access-reviews + audit extras	Privileged Access Review & Additional Stories · Phase 2 High

Appendix F — Complete Audit Event Matrix (75+ Events)

All audit events across all 14 modules. All entries are append-only, SHA-256 row-hashed, actor=SYSTEM for automated events.

AUDIT EVENT	MODULE	KEY FIELDS + REGULATORY REFERENCE
USER_CREATED	User Mgmt	user_id, email, dept, site, role, created_by_id · §11.10(g)
BULK_USER_IMPORT	User Mgmt	filename, file_hash, record_count, success/failure counts · §11.10(g)
USER_ROLE_ASSIGNED	User Mgmt	user_id, role_id, reason, assigned_by_id, approved_by_id · §11.10(g)
USER_DEACTIVATED	User Mgmt	user_id, reason, sessions_revoked, enrollments_suspended · §11.10(g)
USER_REACTIVATED	User Mgmt	user_id, reactivated_by_id, reason · §11.10(g)
SSO_CONFIG_UPDATED	User Mgmt	provider_type, changed_by_id · §11.300
SECURITY_POLICY_UPDATED	User Mgmt	old_value JSONB, new_value JSONB, changed_by_id · §11.300

AUDIT EVENT	MODULE	KEY FIELDS + REGULATORY REFERENCE
ACCESS_REVIEW_COMPLETED	User Mgmt	recertified_count, revoked_count, signed_at · §11.10(g)
COURSE_CREATED	Course Mgmt	course_id, title, gmp_category, created_by_id · Annex 11 §10
SCORM_UPLOADED	Course Mgmt	course_id, file_hash, scorm_version, validation_result · Annex 11
COURSE_APPROVED	Course Mgmt	course_id, workflow_id, all_stage_decisions JSONB, approved_by_id
COURSE_RETIRED	Course Mgmt	course_id, retirement_reason, retired_by_id · Annex 11
COURSE_VERSION_SUPERSEDED	Course Mgmt	old_course_id, new_course_id, superseded_at · Annex 11
ENROLLMENT_CREATED	Enrollment	user_id, course_version_id, due_date, mandatory, created_by_id · §11.10(e)
BULK_ENROLLMENT_IMPORTED	Enrollment	filename, file_hash, created_count, updated_count, error_count · §11.10(e)
AUTO_ENROLLMENT_TRIGGERED	Enrollment	rule_id, user_id, trigger_type, enrollments_created (actor=SYSTEM) · §11.10(e)
ENROLLMENT_MODIFIED	Enrollment	enrollment_id, changed_fields[], old/new values, reason · ALCOA+
ENROLLMENT_CANCELLED	Enrollment	enrollment_id, cancellation_reason, cancelled_by_id · §11.10(e)
RE_ENROLLMENT_GRANTED	Enrollment	original_id, new_enrollment_id, reason, granted_by_id · §11.10(e)
TRANSCRIPT_EXPORTED	Enrollment	target_user_id, record_count, file_hash · §11.10(b)
BATCH_CREATED	Batch Mgmt	batch_id, name, course_count, audience_type · Annex 11 §4
BATCH_ACTIVATED	Batch Mgmt	batch_id, enrollments_created, notifications_sent · Annex 11 §4
BATCH_MEMBER_ADDED	Batch Mgmt	batch_id, user_id, added_by_id · Annex 11 §4
BATCH_MEMBER_REMOVED	Batch Mgmt	batch_id, user_id, removed_by_id, reason · Annex 11 §4
BATCH_REPORT_GENERATED	Batch Mgmt	batch_id, record_count, file_hash · Annex 11 §4
JOB_SPEC_PUBLISHED	Job Specs	spec_id, version, approved_by_id · ICH Q10
JOB_SPEC_ASSIGNED_TO_USER	Job Specs	spec_id, user_id, enrollments_created · §11.10(e)
TRAINING_MATRIX_UPDATED	Job Specs	changed_cells JSONB, qa_esignature_id · ICH Q10 + §11.50

AUDIT EVENT	MODULE	KEY FIELDS + REGULATORY REFERENCE
GAP_ANALYSIS_RUN	Job Specs	spec_id, gaps_found, enrollments_created · Annex 11 §9
JOB_SPEC_NEW_VERSION	Job Specs	old_version, new_version, change_reason · ICH Q10
QUESTION_APPROVED	Assessment	question_id, version, approved_by_id · §11.10(e)
ASSESSMENT_OVERRIDE_APPLIED	Assessment	override_id, original_attempt_id, both_signatories · §11.10(e), ALCOA+
ASSESSMENT_MIN_TIME_VIOLATION	Assessment	attempt_id, time_elapsed_seconds, min_required · §11.10(e)
CERTIFICATE_ISSUED	Cert Mgmt	cert_id, conditions_met JSONB · §11.50
CERTIFICATE_MANUALLY_ISSUED	Cert Mgmt	cert_id, justification, issuance_type=MANUAL, approved_by_id · §11.50
CERTIFICATE_REVOKED	Cert Mgmt	cert_id, reason, esignature_id, both_signatories · §11.50
CERTIFICATE_REGISTER_EXPORTED	Cert Mgmt	filter_params, row_count, file_hash · §11.10(e)
DOCUMENT_UPLOADED	Doc Mgmt	sop_version_id, file_hash, uploaded_by_id · Annex 11 §14
DOCUMENT_APPROVED	Doc Mgmt	sop_version_id, all_stage_decisions JSONB · Annex 11 §14
DOCUMENT_ASSIGNED	Doc Mgmt	sop_version_id, departments[], requires_ack · Annex 11 §14
DOCUMENT_SUPERSEDED	Doc Mgmt	old_version_id, new_version_id · Annex 11 §14
DOCUMENT_ARCHIVED	Doc Mgmt	sop_version_id, retention_policy, e_signature_id · §11.10(c)
GAP_REPORT_GENERATED	Compliance	spec_id, gaps_found, filter_params · Annex 11 §9
REGULATORY_REPORT_GENERATED	Compliance	report_id, scope, e_signature_id · §11.50
TRACEABILITY_MATRIX_EXPORTED	Compliance	filter_params, row_count, file_hash · Annex 11 §9
COMPLIANCE_REPORT_GENERATED	Compliance	spec_id/dept_id, filter_params, file_hash · Annex 11 §9
AUDIT_TRAIL_SEARCHED	Audit Trail	filter_params, result_count, searched_by_id · §11.10(e)
INTEGRITY_CHECK_RUN	Audit Trail	records_checked, violations_found, result · §11.10(e)
CAPA_RAISED	CAPA Mgmt	capa_id, title, owner, linked_courses · ICH Q10

AUDIT EVENT	MODULE	KEY FIELDS + REGULATORY REFERENCE
CAPA_CLOSED	CAPA Mgmt	capa_id, evidence_s3_key, esignature_id · ICH Q10 + §11.50
ESIGNATURE_INTEGRITY_CHECKED	Audit Trail	signature_id, result:valid tampered, checked_by_id · §11.50
INSPECTION_PACKAGE_GENERATED	Audit Trail	package_id, scope, e_signature_id, file_hash · §11.50
NOTIFICATION_TEMPLATE_UPDATED	Notifications	template_type, changed_by_id
ANNOUNCEMENT_SENT	Notifications	announcement_id, recipient_count, scope, sent_by_id
SYSTEM_CONFIG_UPDATED	System Config	key, old_value, new_value, changed_by_id · §11.30
HR_SYNC_COMPLETED	System Config	records_received, processed, failed (actor=SYSTEM) · §11.10(g)
API_KEY_CREATED	System Config	key_name, scopes, created_by_id · §11.30
API_KEY_REVOKED	System Config	key_name, revoked_by_id, reason · §11.30
BACKUP_COMPLETED	System Config	s3_key, file_hash, size_bytes (actor=SYSTEM) · §11.10(c)
VALIDATION_DOCUMENT_UPLOADED	System Config	doc_type, version, file_hash · GAMP 5 §10
RECORDS_ARCHIVED	Data Governance	entity_type, record_count, e_signature_id · §11.10(c), Annex 11 §17
GDPR_ERASURE_EXECUTED	Data Governance	user_id_hash, fields_pseudonymised_count · GDPR Art.17
DATA_EXPORT_GENERATED	Data Governance	target_user_id, record_count, file_hash · GDPR Art.20